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Application of Natural Language Processing Techniques for Detecting Greenwashing in Annual Reports of the Real Estate, Construction, and Materials Industries in Vietnam

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Abstract

Greenwashing is by definition a gap between what a company says and how it behaves, yet existing NLP systems for detecting it read only the company's own reports, carry no explicit notion of time, and are built for English-language Western disclosure. This thesis constructs a temporal ESG knowledge graph for Vietnamese real-estate, construction and building-materials issuers. The graph carries two independent evidence channels, claims extracted from annual reports and conduct evidence extracted from independent news, held in one schema and one identity space but kept separable by a source-type stamp, with sentence-level provenance preserved end to end and a dual-standard axis linking Circular 96/2020/TT-BTC to the GRI Standards. Each claim is adjudicated against the conduct evidence retrieved for it as supported, contradicted or unverified. No greenwashing score is emitted, since no ground-truth label exists against which one could be validated.

Two scopes are reported separately. A sector-wide corpus of 1,416 annual-report filings from 115 listed issuers was crawled, of which 1,216 were parsed and ESG-classified (873,756 sentences), alongside 3,797 distinct news articles from 716 Vietnamese online outlets; both stand as a resource in their own right. The framework itself was then instantiated end to end on five issuers, yielding a resolved graph of 10,624 nodes and 15,130 edges at 100% schema compliance and 97.78% round-trip grounding of extracted values to their cited page. Cross-checking those issuers' 464 claims returns 448 (96.55%) unverified, the direct consequence of a conduct channel roughly sixteen times smaller than the claim side, a shortfall the design reports as abstention instead of forcing a verdict. Only 16 of the 464 claims receive a supported or contradicted verdict at all, and all but one of them belong to a single issuer, ACG. The graph-native indicator-axis path contributed none of the 401 adjudicated pairs, so retrieval in the system as evaluated here is Vietnamese token overlap with a temporal window; it is not yet the graph-structured join the design also specifies. Of the 43 (claim, evidence) pairs the system currently cites, every one carries blind labels from two external domain experts, who confirm 62.8% and 65.1% of them respectively and disagree with one another on a single pair. That result is concentrated in the same issuer, and should not be read as general across all five. Scaling the graph beyond five issuers is engineering already specified by the pipeline, not an open research question.

Keywords: greenwashing detection; temporal knowledge graph; ESG disclosure; Vietnamese NLP; claim verification; label-free evaluation.

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List of Abbreviations and Acronyms

Acronym	**Definition**
AC1	Gwet's AC1, a chance-corrected inter-annotator agreement coefficient
AGM	Annual General Meeting (of shareholders)
AI	Artificial Intelligence
API	Application Programming Interface
BCTN	*Báo cáo thường niên* (annual report), the document class every crawled filing belongs to
BERT	Bidirectional Encoder Representations from Transformers
BTC	*Bộ Tài chính* (Ministry of Finance), as in Circular 96/2020/TT-BTC
BXD	*Bộ Xây dựng* (Ministry of Construction), as in QCVN 09:2013/BXD
CI	Confidence Interval
COP26	26th UN Climate Change Conference of the Parties (Glasgow, 2021)
CSR	Corporate Social Responsibility
EDGE	Excellence in Design for Greater Efficiencies, an IFC green-building certification
ESG	Environmental, Social and Governance
F1	F1 score, the harmonic mean of precision and recall
GHG	Greenhouse Gas
GRI	Global Reporting Initiative
HNX	Hanoi Stock Exchange
HOSE	Ho Chi Minh Stock Exchange
IFC	International Finance Corporation
IPCC	Intergovernmental Panel on Climate Change
ISO	International Organization for Standardization (here, its date-format standard)
JSC	Joint Stock Company
JSON	JavaScript Object Notation
JSONL	JSON Lines (one JSON record per line)
KG	Knowledge Graph
KPI	Key Performance Indicator
LEED	Leadership in Energy and Environmental Design, a green-building certification

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Acronym	Definition
LLM	Large Language Model
LOTUS	Vietnam Green Building Council's green-building certification programme
NFC	Unicode Normalization Form C
NLP	Natural Language Processing
OCR	Optical Character Recognition
OTC	Over-the-Counter (market)
PDF	Portable Document Format
QCVN	*Quy chuẩn Việt Nam* (Vietnamese National Technical Regulation)
QĐ	*Quyết định* (Decision), as in Decision 2171/QĐ-BTC
RAG	Retrieval-Augmented Generation
RSS	Really Simple Syndication
SHA-256	Secure Hash Algorithm, 256-bit (used for source-file provenance)
SSC	State Securities Commission (of Vietnam)
T1	Identity tier of the temporal knowledge-graph schema: timeless entities (e.g. Organization, Facility)
T2	Event/Observation tier of the temporal knowledge-graph schema: dated occurrences (e.g. KPIObservation, Penalty)
T3	Assertion tier of the temporal knowledge-graph schema: claims a source makes (e.g. SustainabilityClaim, Goal)
TCFD	Task Force on Climate-related Financial Disclosures
TF-IDF	Term Frequency–Inverse Document Frequency
TT	*Thông tư* (Circular), as in Circular 96/2020/TT-BTC
TT96	Shorthand used throughout for Circular 96/2020/TT-BTC
TTg	*Thủ tướng Chính phủ* (Prime Minister/Government), as in Decision 13/2024/QĐ-TTg
UI	User Interface
UPCoM	Unlisted Public Company Market
URL	Uniform Resource Locator
VLXD	*Vật liệu xây dựng* (building materials)
VN	Vietnam / Vietnamese
VND	Vietnamese Dong

1. Project Introduction

1.1. Problem and Motivation

The global rise in corporate sustainability reporting has brought a parallel rise in greenwashing, the practice of making misleading or unsubstantiated environmental claims to appear greener than actual conduct warrants [1]. It erodes trust in environmental markets, misdirects capital away from genuinely sustainable firms, and undermines climate goals. As AI systems increasingly inform investment and regulatory decisions, tools able to detect deceptive sustainability communication at scale are becoming urgent rather than merely useful.

Vietnam is a sharp case in point. After committing at COP26 (2021) to net-zero by 2050, the government substantially strengthened its ESG regulatory framework. Circular 96/2020/TT-BTC requires every listed company to disclose environmental and social impacts, and Decision 13/2024/QĐ-TTg extended greenhouse-gas inventory requirements to over 2,166 facilities. Disclosure quality has not kept pace with the regulation, though: only about 25% of companies in Vietnam's top sustainability indices publish independently verified reports, and no legal mechanism specifically targets greenwashing in corporate disclosure.

The gap is especially acute in real estate, construction and building materials, sectors central both to Vietnam's growth and to its emissions footprint. Cement, steel and clay brick carry heavy lifecycle carbon costs. Rapid urbanisation, a growing green-building market and investor pressure together create strong incentives to project green credentials whether or not those credentials are substantiated. Construction alone accounts for 229 of the facilities under Vietnam's mandatory GHG inventory, and yet no automated mechanism currently assesses whether these companies' environmental claims are true.

The sector also happens to be a good methodological fit, not only an important one. Its ESG claims are checkable: energy, emissions, water, waste and recycled-material content are first-order to its operations, it is governed by the national technical standard QCVN 09 on energy-efficient buildings, and its claims resolve into physical quantities that leave traces outside the company's own report. This is unlike the largely qualitative, hard-to-verify disclosure typical of service sectors. A sector whose claims are falsifiable is close to a precondition for studying this problem at all.

Manual detection does not scale, and that has motivated a growing body of NLP research on sustainability communication: sentiment analysis, topic modelling, BERT-based classifiers, and more recently retrieval-augmented generation (RAG) for identifying misleading claims. The most relevant advance is EmeraldMind [2], the first end-to-end knowledge-graph-augmented RAG pipeline for greenwashing detection. It grounds LLM-based claim assessment in a company-level ESG knowledge graph plus a vectorised document store, producing transparent, fact-backed verdicts without fine-tuning, and it outperforms generic LLM baselines on a curated 620-claim benchmark.

Every existing framework, EmeraldMind included, has been built and evaluated on English-language reports from Western regulatory contexts. No prior study applies NLP-based greenwashing detection to Vietnamese annual reports, which are written in Vietnamese, structured by Circular 96 rather than published as standalone sustainability reports, and framed by a national indicator vocabulary that a GRI-only system will not recognise.

A more fundamental limitation cuts across this literature, EmeraldMind included, and it is what this thesis treats as central. A company authors its own sustainability narrative and is also the party that benefits from that narrative being believed. A system whose only input is the corporate report, however sophisticated, inherits the bias it was built to detect: it can judge whether a report is coherent or comprehensive, but not whether it is true. RAG does not close this gap either [3], because greenwashing is not really a retrieval problem. It is a discrepancy between two sources at two points in time. Detecting it needs an evidence channel independent of the company being assessed, a representation that carries time explicitly so a commitment made in 2012 can be compared with conduct observed in 2024, and sentence-level provenance so a reader can open the source and overturn the system's conclusion. That combination points toward a temporal knowledge graph rather than a vector index.

This thesis adapts EmeraldMind's RAG-plus-knowledge-graph paradigm to the Vietnamese context and extends it with a second, independent evidence channel: report-derived claims and news-derived conduct evidence, extracted into one schema and one identity space but kept distinguishable at query time by a channel stamp. Each claim is then matched against whatever conduct evidence is available for it and adjudicated into a provenance-carrying advisory dossier. As far as we can tell, no prior study fuses an independent claim channel and conduct channel into a single temporal ESG graph, in Vietnamese or otherwise.

One consequence follows directly from this design. The system stops short of emitting a greenwashing score or verdict, because no ground-truth label exists for this corpus, or for any Vietnamese corpus, against which such a score could be validated. It returns the evidence instead, an advisory assessment and a full provenance trail, and leaves the judgement to the reader.

1.2. Related Work

1.2.1. Conceptual foundations of greenwashing

Lyon and Maxwell [1] formalised greenwashing as the strategic use of selective environmental disclosure to create a misleadingly favourable image. Delmas and Burbano [4] refined this as the intersection of poor environmental performance and positive environmental communication, distinct from sincere but ineffective sustainability efforts. Regulators such as the European Securities and Markets Authority have since broadened the definition to cover vague and unverifiable claims, not only false ones, which makes automated detection harder still. The practice is enabled by weak accountability: sustainability reporting is largely unaudited, so companies face strong incentives to make ambitious claims and little pressure to substantiate them.

The Delmas–Burbano definition matters here because it more or less dictates an architecture. If greenwashing is the intersection of what a company says and how it behaves, a system with access to only one side cannot in principle observe it. The related work below is assessed against that requirement, family by family.

1.2.2. NLP for ESG and sustainability report analysis

Early work on extracting sustainability information from corporate disclosures relied on keyword dictionaries and term-frequency methods, with limited ability to handle contextual nuance. Transformer-based models changed this. ClimateBERT [5], pretrained on over two million climate-related paragraphs, became a foundational tool for classifying climate-relevant content, and domain-specific models such as FinBERT-ESG and ESG-BERT extended this across the E/S/G dimensions.[1]

Beyond classification, NLP has been applied to topic detection, specificity and readability scoring, question answering over climate reports, and structured data extraction from ESG documents with LLMs. Schimanski et al. [6] extracted nature-related topics such as water, forests and biodiversity from annual and sustainability reports, showing that fine-grained domain analysis is feasible on corporate reporting corpora. This body of work establishes that NLP can surface meaningful sustainability signals from complex annual-report text, and it is the methodological foundation greenwashing-detection approaches build on.

1.2.3. Automated greenwashing detection

Automated greenwashing detection has attracted growing attention as mandatory disclosure has expanded globally. Gorovaia et al. [7] found that companies with environmental violations produce longer, more positively worded, less readable CSR reports, consistent with obscuring poor performance through narrative complexity. Bingler et al. [8] reached a related conclusion, showing that TCFD-supporting firms produce climate disclosures that are structurally compliant but substantively vague. Both studies infer greenwashing from properties of the text alone.

Two other groups measure disclosure against an external reference instead. Kim et al. [9] built a BERT-based greenwashing sentence-scoring model for a public monitoring service that tracks companies from media data, and a study of Central and Eastern European firms [10] built a Greenwashing Severity Index from sentiment analysis, TF-IDF and topic modelling to quantify the gap between ESG self-reports and media narratives, finding moderate but widespread greenwashing. Vinella et al. [11] fine-tuned ClimateBERT against a purpose-built greenwashing-risk formulation, reporting 86.34% accuracy and an F1 of 0.67.

Read against the Delmas–Burbano criterion, these fall into roughly three groups. Disclosure-only approaches [7, 8] measure text properties alone, so they detect rhetorical style rather than factual discrepancy; a well-written false claim evades them entirely. Claim-versus-rating approaches compare a firm's disclosure against a third-party rating, but the rating is itself largely derived from disclosure, so the comparison ends up circular. Only claim-versus-independent-outcome approaches actually observe both halves of the intersection, and they are rarest because

[1] FinBERT-ESG and ESG-BERT are publicly released checkpoints in active use (Hugging Face: `yiyanghkust/finbert-esg`, `nbroad/ESG-BERT`), but neither has a stable peer-reviewed paper to cite as of this writing; they are named here as artefacts, not as citable studies.

the independent half is costly to obtain. The Central and Eastern European study [10] comes closest.

A further constraint matters for how this thesis is evaluated. The first comprehensive NLP survey on greenwashing detection [12], reviewing 61 studies, identified three persistent challenges: no standardised definition, scarce reliably labelled data, and poor generalisation of English/Western-trained models to other contexts. The first is why this work returns advisory evidence rather than a classification, with a label-free evaluation protocol; the second is why it targets the Vietnamese regulatory environment directly.

1.2.4. Retrieval-augmented generation and knowledge-graph-augmented approaches

The most directly relevant precedent is EmeraldMind, the first end-to-end knowledge-graph-augmented RAG pipeline for greenwashing detection. It builds two evidence stores, EmeraldGraph (a knowledge graph of company-specific ESG entities and relationships extracted from ESG reports) and EmeraldDB (a vectorised document store), retrieves supporting or contradicting evidence from both for a given claim, and passes it to an LLM that classifies the claim as greenwashing, not greenwashing, or abstains when evidence is inconclusive, with a natural-language justification. It achieves competitive accuracy and markedly better explanation quality than vanilla LLM baselines, without fine-tuning.

Related RAG work in the ESG domain includes ESGReveal [13], which applies RAG to structured ESG data extraction, and ChatClimate [14], which grounds climate question answering in the IPCC's Sixth Assessment Report rather than any single company's disclosures. More broadly, knowledge-graph integration has been shown to improve claim-verification accuracy by supplying structured background knowledge an LLM cannot reliably retrieve from parametric memory alone.

Two properties bound what this line of work can do, and both are addressed directly here. Its evidence stores are built entirely from the company's own reports, so retrieval returns only what the company said elsewhere in its own documents. That is a strong consistency check, but by the Delmas–Burbano criterion it is still only the communication half of the intersection. The representation is also atemporal: published ESG knowledge graphs, EmeraldGraph included, model entities and relations without validity intervals, so they cannot express that a company committed to X in 2021 and acted contrary to X in 2024, which is the exact shape of the phenomenon under study. The temporal-database literature distinguishes valid time (when a fact holds) from transaction time (when it was recorded), and temporal-KG work extends entity representations with validity intervals and version chains [15], but that line has not yet been combined with corporate disclosure analysis.

1.2.5. ESG disclosure and greenwashing in Vietnam

The Vietnamese corporate reporting context remains almost entirely unstudied computationally, despite a fast-evolving disclosure framework. Circular 96/2020/TT-BTC mandates environmental and social disclosure for all listed companies, the State Securities Commission's ESG Implementation and Disclosure Handbook (2024) offers GRI-aligned guidance, and the

Vietnam Sustainability Index (2017) benchmarks the twenty most sustainable HOSE-listed companies. Independent verification nevertheless remains rare, and greenwashing is a recognised, growing risk.

Real estate and construction sit at the centre of this risk. Both are named major GHG emitters under Vietnam's Emissions Trading Scheme, and the rapid spread of green-building certifications such as LEED, LOTUS and EDGE creates incentives for green branding that may exceed the cost of genuine implementation. No prior computational study has examined whether Vietnamese real-estate, construction and materials companies substantiate the environmental claims in their annual reports.

The Vietnamese setting also imposes a requirement the international literature does not face: disclosure is framed in a national indicator vocabulary (Circular 96/2020/TT-BTC, Decision 2171/QĐ-BTC, QCVN 09:2013/BXD, and the SSC–IFC reporting guide), and none of these maps onto GRI term-for-term. A system that recognises only GRI vocabulary will miss the disclosures Vietnamese regulation actually obliges. This is why the thesis builds a dual-standard indicator axis rather than adopting either standard alone.

1.3. Contribution

This thesis makes the following contributions.

1. A Vietnamese ESG disclosure corpus for greenwashing analysis, built from the annual-report filings of 115 listed companies in the real-estate, construction and building-materials sectors (94.8% filed on the Ho Chi Minh Stock Exchange, the remainder on HNX and UPCoM), comprising 1,416 documents.

2. An independent conduct channel, the property that distinguishes the proposed system from disclosure-only greenwashing detection: 3,797 distinct articles (4,582 documents crawled, before cross-ticker de-duplication) from 716 Vietnamese online outlets, ingested into the same schema and identity space as the reports but kept separable at query time by a `source_type` stamp carried on every node and edge. Claims and conduct can therefore be compared without either being able to masquerade as the other.

3. A temporal ESG knowledge-graph schema adapted to Vietnamese regulatory and linguistic norms, with 28 node classes and 48 edge labels declared over 76 legal class pairs, governed by eight explicit design principles and a machine-checkable invariant set. The rule that entity identity must be timeless, along with the T1/T2/T3 tier partition, is enforced by an offline linter rather than left to convention, so a violation fails a test instead of surviving review.

4. A dual-standard indicator axis unifying Circular 96/2020/TT-BTC with the GRI Standards through a confirmed-only crosswalk, materialised as graph structure so that a Vietnamese-vocabulary disclosure and its international equivalent resolve to one node.

5. A label-free evaluation instrument, none of whose checks need a ground-truth label to run, demonstrated on three controlled before/after ablations: for instance, the negative control's

same-company evidence-attribution check (§4.3) moved from 28.76% (failing) to 100% (passing, zero cross-feed citations) once the adjudication-cache defect of §3.8 was fixed, showing the instrument can detect a real regression and its repair without any label existing for the underlying task.

6. A documented and defended negative design decision. The pipeline classifies each claim–evidence pair as *supports*, *contradicts* or *irrelevant*, and each claim as *contradicted*, *supported* or *unverified* due to insufficient evidence, but it never emits a greenwashing score or verdict, because no label exists against which such a score could be validated.

2. Data

2.1. Dataset Overview

Progress in evidence-based greenwashing analysis requires a corpus that meets three conditions at once, and no existing public dataset meets all three. It has to be Vietnamese, since the disclosures the regulation obliges companies to make are written in Vietnamese and framed in a national regulatory vocabulary. It has to be temporally deep, because the phenomenon under study is a discrepancy between a statement made in one year and conduct observed in another; a single-year snapshot cannot express that. And it needs two structurally independent channels, since a corpus built only from company-authored reports can support consistency checking but not verification.

Existing ESG corpora each satisfy one or two of these conditions and fall short on the rest. International sustainability-report collections are temporally deep but monolingual in English and single-channel. Vietnamese financial-text datasets are in the right language but built for sentiment or event extraction, with no regulatory indicator vocabulary attached. News corpora provide independence but no claims to check against. The corpus described in this chapter was assembled to close that gap.

2.1.1. Unit of analysis and record schema

The dataset's unit of analysis is a single sentence, carried with the coordinates that locate it in a published document. A document-level record would not do the job here: a company does not greenwash a report as a whole, it makes an individual assertion on a page, and that assertion is what has to be checkable. Every record is a JSON object keyed by the triple (source pdf, page, sentence index), assigned once when the sentence is first segmented and never rewritten afterward.

Table 2.1: Core record-schema fields, common to both channels (full schema in Appendix A).

Field	Type	Meaning
`source_pdf`	string	Source document identifier (report file name, or ticker + domain + hash for an article)
`page`	int	1-based page of the source PDF (always 1 for an article)
`sentence_index`	int	Position of the sentence within that page
`text`	string	The sentence, verbatim, NFC-normalised
`labels`	list	Pillars whose score reaches the tag threshold; may be empty or hold several

Table 2.1 is what makes traceability concrete. The first three fields are assigned once and preserved through classification, extraction, validation, entity resolution and graph loading. Any statement the finished system makes can therefore be resolved back to one sentence, on one page, of one document.

2.1.2. Primary data and reference data

The dataset has two parts, and how each is acquired differs enough to keep them separate. Primary data is the evidence the system reasons over. It is crawled, it grows on each re-run, and it is what the exploratory analysis later in this chapter characterises. Reference data is the controlled vocabulary the system reasons with. It is built once from published standards, it changes only when those standards change, and it is committed to version control alongside the source PDFs and their SHA-256 digests.

Table 2.2: The dataset, measured from the named artefacts.

Part	Component	Source	Scale
Primary	Raw report corpus	Annual-report filings, 94.8% HOSE, via VietStock's disclosure mirror	1,416 PDFs, 115 issuers, 16.93 GB
Primary	Labelled report sentences	The above, parsed and ESG-classified	873,756 sentences, 1,216 documents
Primary	News corpus	716 Vietnamese online outlets, three search back-ends	3,797 articles, 174,256 sentences
Reference	VN regulatory corpus	TT96/2020, QĐ 2171, QCVN 09, SSC–IFC	35 KPI definitions
Reference	GRI standards corpus	GRI (official PDFs)	42 PDFs → 136 disclosure codes
Reference	Crosswalk	Hand-curated, confirmed rows only	20 GRI codes with a TT96 equivalent

Table 2.2 shows the corpus separating breadth from depth. The raw pool and the labelled sentence corpus span all 115 issuers across the sector. End-to-end graph construction and cross-check, in contrast, run in depth on a single issuer with a fifteen-year time series. This is a deliberate trade-off, and its consequence for the strength of the claims made here is taken up later in this thesis. Two boundaries keep the evaluation in Chapter 4 non-circular. First, the KPI records, the extracted triples and the resolved knowledge graph are outputs of the system under study. They were not handed to it as input data; they are produced by the stages described in Chapter 3 and reported as results in the Experiment chapter, so they are not listed here as corpus components. Second, there is no train/validation/test split, because no component of the system is trained. The ESG classifier is a published checkpoint applied as released (§2.2.4), and every other stage is a rule, an offline algorithm, or a prompted model used without gradient updates. The labelled sample a proper classifier evaluation would require does not yet exist and is recorded as an outstanding item.

2.2. Data Construction Pipeline

2.2.1. Pipeline Overview

This chapter's data construction runs as three independent pipelines. A report pipeline downloads published annual-report PDFs and linearises them into sentences. A news pipeline queries three independent search back-ends and extracts article bodies into the same sentence schema, so both channels can later share one identity space while staying distinguishable by a source-type stamp. The two run separately and converge through one shared ESG classifier (Figure 2.1, drawn once since it is the same weights applied to each), then diverge again into the claim corpus and the conduct corpus that feed graph construction (Chapter 3).

A third pipeline runs on its own schedule and does not depend on the first two. A pair of run-once paths turns published regulatory and GRI text into the controlled vocabulary: a 35-indicator KPI vocabulary and a 136-code GRI catalogue (Figure 2.2, §2.2.6), consumed downstream by KPI extraction and the indicator axis. The traceability guarantee that runs through all three pipelines (each record keyed by source pdf, page and sentence index, §2.1.1) holds throughout and is not re-derived per stage.

2.2.2. Public data sources and the raw data pool

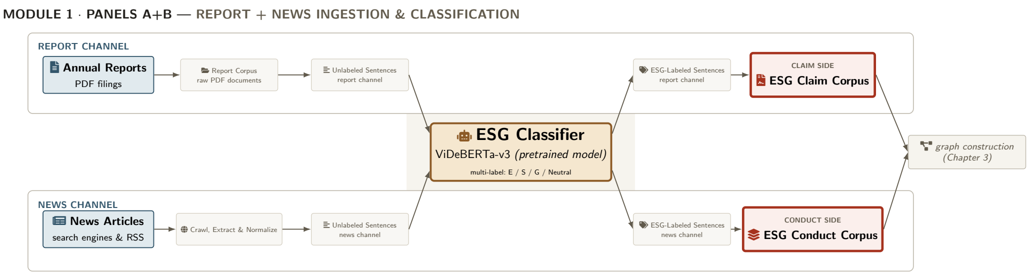

Figure 2.1: Report and news ingestion and classification pipeline.

Vietnamese listed companies must file an annual report with the exchange they are listed on, which then publishes the filing. The corpus is therefore built from exchange disclosure filings, not from company websites. A filing is the document the issuer is legally answerable for, it is archived at a stable location, and it is available uniformly for every ticker. An investor-relations page, by comparison, is voluntary and inconsistently maintained.

In practice the filings come from VietStock's static mirror of the disclosure archive, not from each exchange's own portal. That mirror exposes the complete filing history of every ticker under one stable, machine-readable URL scheme. The scheme encodes the exchange, the document class (BCTN, *báo cáo thường niên*, annual report), the language, and two distinct year fields, one on the directory and one in the file name, so every field the acquisition index needs is recoverable from the address alone. The two year fields are not the same quantity, a point taken up below. The cost of this route is a dependency on a single third-party host that is not the authoritative publisher of the filings, a limitation returned to in §5.3.

Acquisition is a crawl, and the crawl produces an index. Discovery is kept separate from download. A crawl over the sector's ticker list resolves each issuer's filing history into one row per document, and that index, `config/company_annual_report.xlsx`, sheet *Xây dựng - VLXD - BĐS*, carries the ticker, the issuer's legal name, the document class, the filing year and the direct file URL for all 1,357 filings across 115 issuers, all served from a single host. Keeping the index as a materialised artifact instead of re-resolving URLs at download time makes the acquisition auditable and repeatable. The same index is reused as the issuer list for the news channel (§2.2.5) and for the issuer-identity registry.

Table 2.3: The acquisition index, measured from the crawl manifest.

Property	Value
Filings indexed	1,357
Issuers (tickers)	115
Document class	BCTN - annual report (100%)
Listing venue	**HOSE 1,287** / HNX 51 / OTC 11 / UPCOM 7 / KHAC 1
Filing years spanned	2011–2026
File format	pdf 1,259 / rar 49 / zip 47 / doc 2

Table 2.3 settles three things that matter downstream. The corpus is overwhelmingly HOSE, at 94.8% of filings, with 51 HNX and 7 UPCoM rows present because a few issuers moved venue during the period covered; any statement about *listed Vietnamese companies* in this thesis should be read as a statement about HOSE-listed ones. Every indexed document is an annual report, so the ESG content analysed here is ESG content as it appears *inside* a general-purpose annual report, a materially thinner genre than a standalone sustainability report. And 7.1% of filings arrive as `.zip` or `.rar` archives instead of a plain PDF, which is why extracting `.rar` and `.7z` containers is a required step of the downloader.

There is one naming caveat that affects every temporal figure in this thesis. The year in a downloaded file's name is the *filing* year, and for 95.5% of rows it is exactly one greater than the *fiscal* year the report covers: a report on financial year 2024 is filed in spring 2025. The two are not interchangeable, and this thesis uses *filing year* wherever a year is read off a file name, labelling it as such. The distinction shifts the whole temporal axis by one year, and it is why the compliance discontinuity discussed in §2.3.4 shows up in the 2021 filings, not the 2020 ones.

After download and unpacking, the raw pool holds 1,416 PDF files for 115 companies, totalling 16.93 GB at a mean of 12.0 MB per file and 12.3 documents per company, spanning filing years 2011–2026. The file count exceeds the 1,357 indexed filings because an unpacked archive frequently contains several PDFs, typically a report split into chapters. No filtering by content is applied at this stage. Selection happens later, at the sentence level, so the decision to discard is written down explicitly there instead of being left implicit here.

The conduct channel is built out of news, not filings. Per-company queries are generated from the same issuer index and issued against three independent back-ends: Google News RSS, Bing and DuckDuckGo. Article bodies are then fetched, cached and extracted with trafilatura. Using three back-ends instead of one is meant to hedge against any single provider's ranking bias; §2.3.5 reports how unevenly the three actually contributed.

2.2.3. Cleaning, parsing and sentence segmentation

Three cleaning operations are applied, in this order.

The first is text extraction with page anchoring. Report PDFs are converted with a positional text extractor, PyMuPDF[1], which iterates over text runs in coordinate order, normalises to Unicode NFC so Vietnamese diacritics survive, and attaches the page number to every block. This choice is deliberately different from the layout-aware parser used for the standards corpus (§2.2.6), because what this channel needs is an exact page number per sentence, not table reconstruction; that trade-off is argued for here rather than quantified by a head-to-head ablation against the layout-aware parser, which this thesis does not run.

The second is Vietnamese-aware sentence segmentation. Sentences are segmented with underthesea, an open-source Vietnamese NLP toolkit[2], instead of by punctuation rules. Vietnamese financial reporting is dense with decimal separators written as 1.234,5, abbreviations such as TNHH, CP and TP., and full stops inside proper nouns, all of which a naive full-stop rule would fragment.

The third is minimum-length filtering. The splitter discards a segment unless it clears two floors: at least 25 characters and 4 words. Discarding happens here, so the decision is recorded once instead of being re-made implicitly by every later stage. The shortest surviving sentence is exactly 25 characters, the floor itself, so page furniture such as folio numbers and running headers never reaches the classifier. The cost is that a genuinely short disclosure, a bare table cell containing a figure, is lost along with it. That is one reason quantitative values are extracted separately by the KPI-extraction stage (§3.3) instead of being expected to survive as sentences.

2.2.4. Annotation: multi-label ESG classification

Each sentence is annotated with the label set {Environmental, Social, Governance, Neutral} by `nguyen599/ViDeBERTa-v3-ESG-base`, a publicly released multi-label DeBERTa-v2 checkpoint for Vietnamese ESG text, fine-tuned from the ViDeBERTa architecture [16] and distributed on Hugging Face[3]. The model is applied as published. No training happens anywhere in the pipeline; the annotation stage simply runs inference at a maximum sequence length of 256 tokens. Building a labelled Vietnamese ESG training set was outside the project's budget, a deliberate scope decision with a consequence stated in §2.3.3 and revisited in §5.3: the checkpoint was fitted to someone else's data distribution, so how it behaves on this corpus is an empirical question, not a given.

The task is multi-label, not multi-class, because a single sentence can legitimately carry more than one dimension. *"The Board of Directors approved the emission-reduction plan"* is simultaneously a governance statement and an environmental one, and a single label would discard half of it. The model applies a per-label sigmoid instead of a softmax, so the four scores do not sum to one and each is read independently.

[1] `https://pymupdf.readthedocs.io/`
[2] `https://github.com/undertheseanlp/underthesea`
[3] `https://huggingface.co/nguyen599/ViDeBERTa-v3-ESG-base`

Two separate rules produce two different fields, and the distinction matters. A pillar tag is attached when that pillar's sigmoid score reaches 0.45. The binary `esg` flag is decided differently, by the Neutral score falling below 0.5. The second rule stays robust to a signal that splits across pillars: a sentence scoring 0.3/0.3/0.3 on E/S/G is not neutral even though no single pillar clears 0.45. It also means the two fields can and do disagree in both directions, a gap §2.3.3 measures. The model emits the full probability vector alongside both fields, which is what makes that measurement, and the confidence analysis in §2.3.2, possible at all.

The same checkpoint, thresholds and code path are applied to both channels. This is a correctness requirement more than a convenience: if reports and news were labelled by different models, any measured difference between the channels, such as the pillar inversion reported in §2.3.6, would be confounded by the difference between the annotators.

2.2.5. The independent conduct channel

News articles pass through the same sentence schema and classifier as reports, then through a preprocessing stage that normalises publication dates, adds `publish_year` and `date_uncertain` fields, and drops boilerplate. Two contracts on this channel matter more than the pipeline mechanics.

The first is the `date_uncertain` contract. When an article states no explicit date or period for the fact it reports, the system falls back to the publication date as a proxy, but it has to mark that fact `date_uncertain = true`. The flag travels with the evidence into the final dossier as an explicit caveat; the system never silently assumes a year. §2.3.5 shows why this matters: a measurable fraction of crawled articles carry publication dates that are clearly parser artefacts.

The second is independence itself. Articles served from domains owned by the company under assessment are not admissible as independent evidence. They are retained and stay visible to the reader, but they are excluded from the verification path. Without this guard, a company distributing its own press releases widely enough could end up verifying its own claims, which would invert the whole point of the channel.

2.2.6. The reference corpus: regulatory and GRI standards

A third and fourth run-once path build the controlled vocabulary that the graph-construction stages consume, independently of the report and news pipeline in Figure 2.1. One turns four Vietnamese regulatory instruments into a 35-indicator KPI vocabulary; the other turns the GRI Standards into a 136-code catalogue crosswalked to it. Figure 2.2 shows both pipelines in detail.

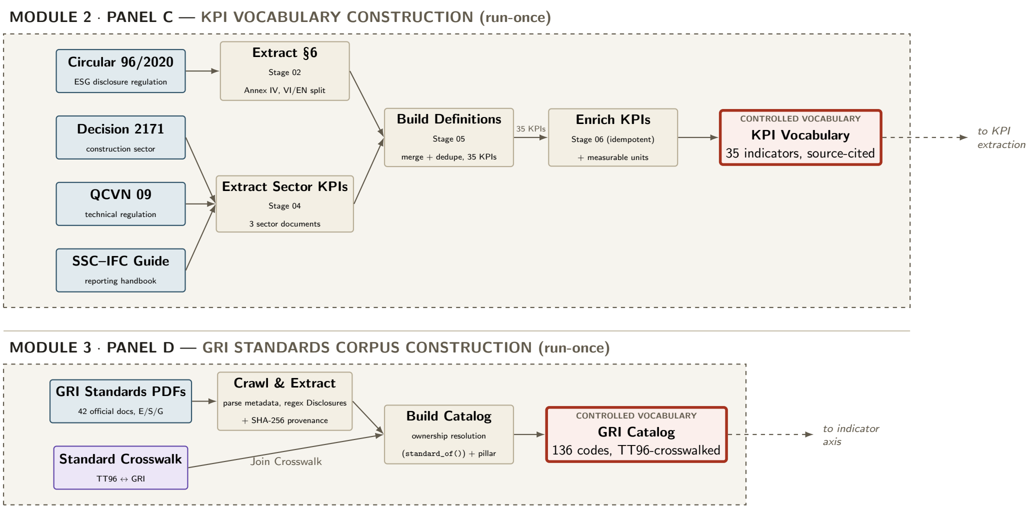

Figure 2.2: Reference-corpus construction: two run-once, offline paths building the controlled vocabulary consumed downstream.

For the KPI vocabulary, four Vietnamese regulatory instruments (Circular 96/2020/TT-BTC, Decision 2171/QĐ-BTC, QCVN 09:2013/BXD and the SSC–IFC ESG reporting guide) are downloaded and their relevant provisions extracted *verbatim* in two parallel passes. Circular 96's Section 6, the annual-report template's ESG-disclosure annex in eight sub-sections, is parsed for its Vietnamese/English indicator pairs, while the three sector-specific sources are parsed for the non-fired building-materials target, the energy-efficiency compliance clause and the SSC–IFC guide's fourteen recommended aspects. The two extractions are merged and deduplicated by indicator id into 35 KPI definitions, each carrying only the source's verbatim wording and a source block naming the instrument, clause and page. A final enrichment pass then attaches a short label and a measurable definition with unit and normalisation hints to every indicator, since the regulations state *topics to disclose*, not *metrics to measure*, and several extracted rows (e.g. "Tái chế") are too thin on their own to drive numeric extraction, while preserving the original verbatim text in a `source.excerpt` field for audit. The pass is idempotent: re-running it re-applies the same curated map without overwriting the captured verbatim.

For the GRI catalogue, 42 official GRI Standards PDFs are linearised into one JSON file per standard, carrying the standard's own metadata (parsed from its filename), its disclosures (extracted by a regular-expression scan for `Disclosure N-n` headings), and a SHA-256 of the source PDF for provenance. Building the catalogue is not a plain merge, though. Several standards, the sector standards GRI 11–14 and the 2024/25 rewrites GRI 101–103, re-list disclosures that belong to other standards, so a disclosure has to be attributed to the one standard whose id is its own prefix, and not to whichever file happens to be read first. This is a documented failure mode: an earlier version that simply read files in sorted filename order mis-attributed 80 of the eventual 136 entries, because `"gri_101"` sorts before `"gri_2"` and so GRI 2-27 was published under GRI 101's title, PDF and version. The corrected pass produces 136 disclosure codes, each joined to a hand-curated TT96 ↔ GRI crosswalk of which only explicitly confirmed rows are used.

This corpus changes only when the standards themselves change, so both paths are built once, committed to version control alongside the source PDFs and their SHA-256 digests, and treated thereafter as generated data. How the two vocabularies join into the graph's indicator axis is described in the Methodology chapter, and its yield is measured directly as coverage later in this thesis (§4.1), not by an ablation.

2.2.7. Final packaging and distribution

Sentences are stored as JSONL, one record per sentence, carrying the traceability triple, the text, the full probability vector, the thresholded labels and channel-specific fields (article URL, domain, publication date and crawl channel for news). Generated data is distributed through a dataset repository instead of Git; the pushed revision is pinned in a version file that *is* tracked in Git, so a checkout recovers the data that went with that code. This is what makes the before/after negative-control check reported later in this thesis (§4.3) reproducible instead of merely asserted.

2.3. Exploratory Data Analysis

This section reports the exploratory analysis of the corpus, placed before the methodology because several design decisions in Chapter 3 respond directly to what is measured here. Every figure and table below is produced by `docs/eda_out/eda_report_data.py`, streaming the corpus directly from disk.

The scope is the data only: raw filings, derived sentences, ESG annotation, temporal spread, conduct-channel composition. What the pipeline subsequently produces from it, such as KPI records, triples and the resolved graph, is a result and not an input (§2.1.2); it is measured later, in the Experiment chapter.

2.3.1. Scale and coverage of the raw corpus

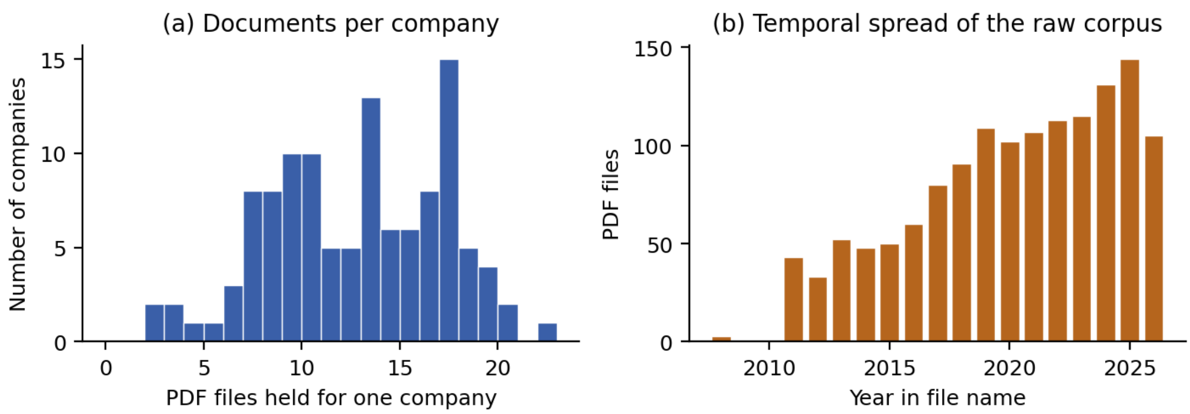

Figure 2.3: Raw report corpus. (a) Distribution of documents held per company. (b) Temporal spread, by the reporting year recoverable from the file name.

Figure 2.3(a) shows coverage per company is uneven but rarely thin, with a mean of 12.3 documents per company across 115 companies. Panel (b) shows the corpus thickening markedly over time. Two causes are confounded here and cannot be separated from file counts alone: listed companies genuinely publishing more, and recent filings simply being easier to retrieve from public sources. Either way the consequence is the same. Evidence density is a function of year, and any comparison across years has to account for it instead of assuming a stationary corpus.

2.3.2. Sentence-level properties

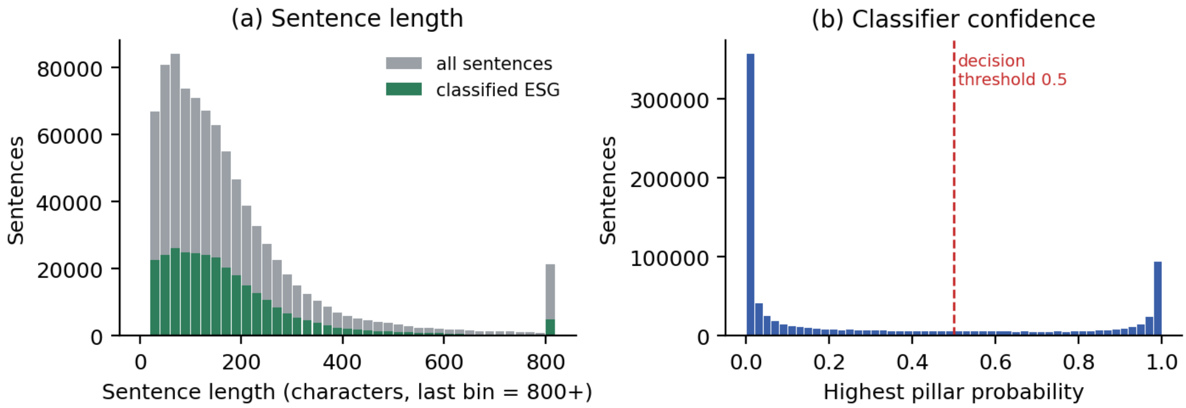

Figure 2.4: Sentence-level properties of the labelled report corpus. (a) Sentence-length distribution, with the ESG-classified subset overlaid. (b) Distribution of the highest pillar probability per sentence, against the 0.5 decision threshold.

Parsing and segmentation produced 873,756 sentences from 1,216 documents spanning 89,731 pages, a mean of 73.8 pages per document. Figure 2.4(a) shows the resulting length distribution. The mean sentence length is 196 characters and the longest is 11,827, which is not really a sentence at all but a table flattened into one text run by the positional extractor, the clearest single artefact of the extraction choice discussed in §2.2.3.

Figure 2.4(b) is the more consequential view. The distribution of the highest pillar probability is strongly bimodal, with mass concentrated near 0 and near 1 and little in between. A classifier whose scores concentrate at the extremes is confident, but confidence and correctness are two different things and the two should not be conflated. What the distribution does constrain is this: the 0.5 threshold is not cutting through a dense region, so small threshold changes will not move the label counts much. It says nothing about whether the confident decisions are right, which only the evaluation table requested in §2.2.4 can answer.

2.3.3. ESG label distribution and a precision concern

Table 2.4 supports four observations, and the last two of them qualify the annotation quality.

The two annotation rules disagree measurably. §2.2.4 explains that a pillar tag needs a sigmoid score of 0.45, while the `esg` flag is set by the Neutral score falling below 0.5, and these different tests diverge in both directions on this corpus: 30,783 sentences (10.1% of the ESG set) are flagged ESG-relevant with no pillar label, and a further 9,255 carry a pillar label but are not flagged ESG-relevant. Neither is a bug. The first is intended behaviour for a signal that splits across pillars, the second for a sentence that is confidently neutral yet mentions one pillar in

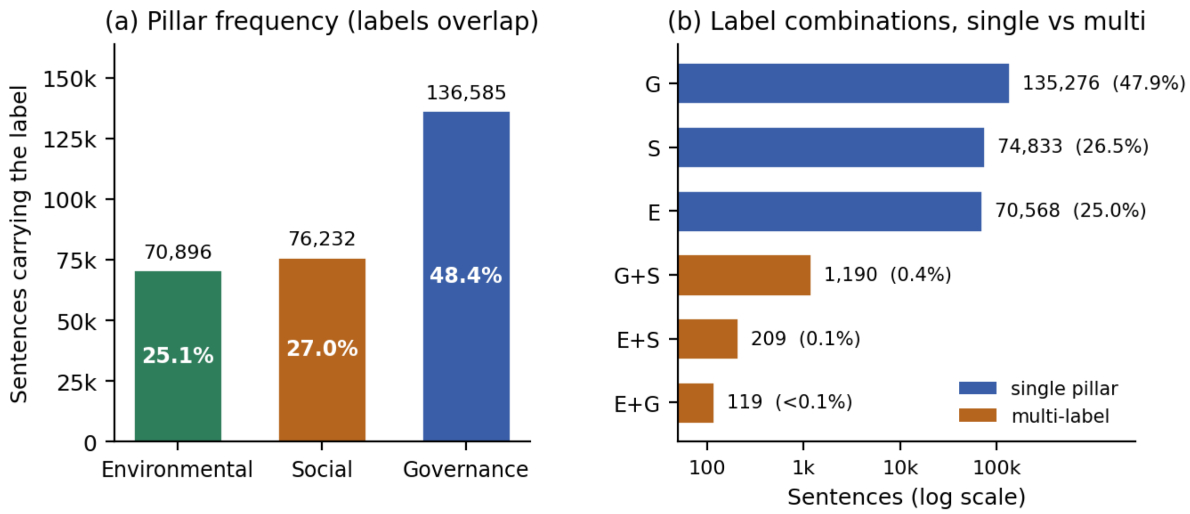

Figure 2.5: ESG annotation of the report corpus, scoped to the 282,195 sentences that carry at least one pillar label. (a) Frequency of each pillar; a sentence may carry several, so the bars do not sum to the total. (b) Every label combination that occurs, single-pillar against multi-label, on a log axis. The unlabelled remainder is excluded from both panels (Table 2.4).

Table 2.4: ESG annotation of the report corpus. Pillar shares are taken over the pillar-labelled subset, not the whole corpus.

Quantity	Sentences	Share
Total sentences	873,756	100% of corpus
Classified ESG-relevant (`esg` flag)	**303,723**	**34.8% of corpus**
Carrying ≥1 pillar label	**282,195**	**32.3% of corpus**
- Governance	136,585	48.4% of labelled
- Social	76,232	27.0% of labelled
- Environmental	70,896	25.1% of labelled
Carrying more than one label	1,518	0.5% of labelled

passing. But the two fields should never be used interchangeably. The pipeline respects this: page selection for the expensive extraction stage keys on the `esg` flag (§3.3), while the evidence interface's pillar columns are read from the indicator axis instead of these labels, as described in the Evidence View UI design later in this thesis.

A second pattern is the imbalance between pillars: Governance appears on 136,585 sentences against 76,232 for Social and 70,896 for Environmental, roughly 1.9× the environmental count. This is consistent with how a Vietnamese annual report is structured: board composition, shareholder structure, internal control and related-party transactions are mandatory, lengthy sections, while environmental disclosure is comparatively short. The downstream effect is direct. The claim side of the graph ends up governance-heavy while the conduct side, as §2.3.6 shows, is environmental-heavy, so the two channels do not measure the same thing in the same proportion.

The multi-label formulation, meanwhile, is barely exercised in practice. Panel (b) of Figure 2.5 makes this visible: on a log axis the three single-pillar bars sit at 10^4–10^5 while all three multi-label bars sit below 1,200. Only 1,518 sentences, 0.5% of the pillar-labelled subset, carry more than one pillar, and Governance + Social (1,190) accounts for 78% of even that. Environmental + Governance, the combination one would most expect in a disclosure about board-approved emission targets, occurs 119 times in nearly nine hundred thousand sentences. The model behaves, in practice, as a single-label classifier. That does not invalidate the multi-label design, which is correct in principle and costless; it simply means it is not currently exercised, and no stronger claim is made for it here.

Finally, the ESG rate of 34.8% looks implausibly high. Roughly one sentence in three of an entire annual report being ESG-relevant does not match these documents' composition on inspection. The likely mechanism is lexical: Vietnamese corporate language uses environmental and social vocabulary in contexts that are not disclosures at all. A concrete instance is checkable in this corpus: the legal name of one issuer is *Công ty Cổ phần Nhựa và Môi trường Xanh An Phát*, literally *An Phat Green Environment and Plastics JSC*. The string *môi trường xanh* ("green environment") appears in 655 sentences, 189 of them (28.9%) labelled Environmental, including bare title-page lines stating nothing but the company's name.

That pattern alone does not account for the whole gap; 189 sentences out of 303,723 is negligible in total. But it establishes the mechanism. A company name, slogan or department title can trigger a pillar label on lexical grounds alone, and there is no particular reason that mechanism is confined to one issuer's name. Quantifying its full extent requires the labelled evaluation set requested in §2.2.4. Until then, the ESG rate reported here should be read as an upper bound on ESG-relevant content, and the downstream stages are designed on that assumption. §3.3 describes the response: the extraction stage never treats an ESG label as evidence of anything, only as a cost-saving filter over which pages are worth sending to an expensive model.

2.3.4. Temporal coverage

Figure 2.6 and Table 2.5 plot this by reporting year. The corpus spans 2008–2026, sentence volume rises roughly fourfold across the period, and the ESG share rises with it, from 24.3% to 38.7%. The rise is gradual. That is worth noting because Circular 96/2020 took effect in 2020,

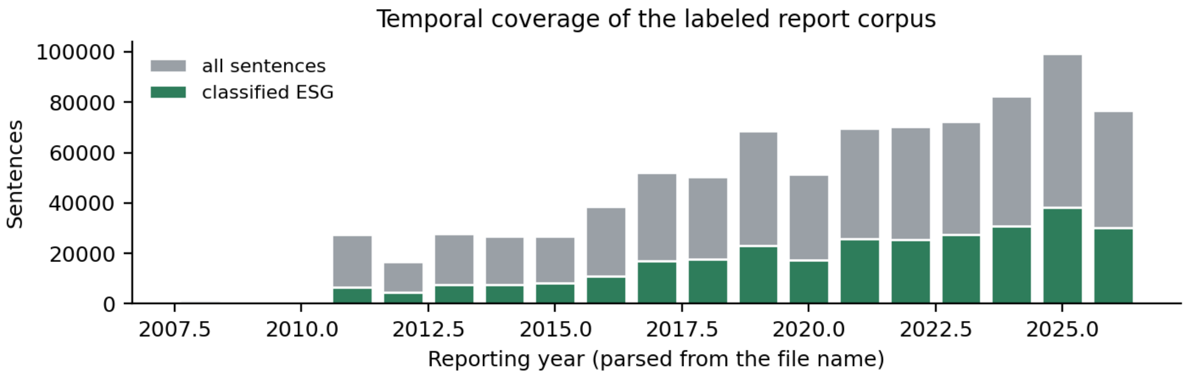

Figure 2.6: Temporal coverage of the labelled report corpus, by reporting year, with the ESG-classified subset overlaid.

Table 2.5: Temporal coverage of the labelled report corpus (odd years shown; all years appear in Figure 2.6).

Reporting year	Sentences	ESG sentences	ESG share
2011	27,318	6,641	24.3%
2013	27,834	7,588	27.3%
2015	26,889	8,369	31.1%
2017	52,215	17,093	32.7%
2019	68,579	23,048	33.6%
2021	69,678	25,877	37.1%
2023	72,222	27,518	38.1%
2025	99,172	38,331	38.7%

and a disclosure mandate that bound immediately should show up as a discontinuity in that year. It does not.

Three explanations are consistent with a gradual rise, and the corpus cannot distinguish between them: genuine secular growth in ESG disclosure independent of the mandate, a gradual compliance ramp-up instead of immediate adoption, or drift in the vocabulary the classifier keys on with no real change in underlying content. The third would just be a measurement artefact dressed up as a finding, and it cannot be excluded without the evaluation set of §2.2.4. The claim made here is restricted to what the data supports: ESG-classified content grows as a share of the corpus over time, and that growth is not attributable to the 2020 mandate on the evidence available.

2.3.5. The independent conduct channel

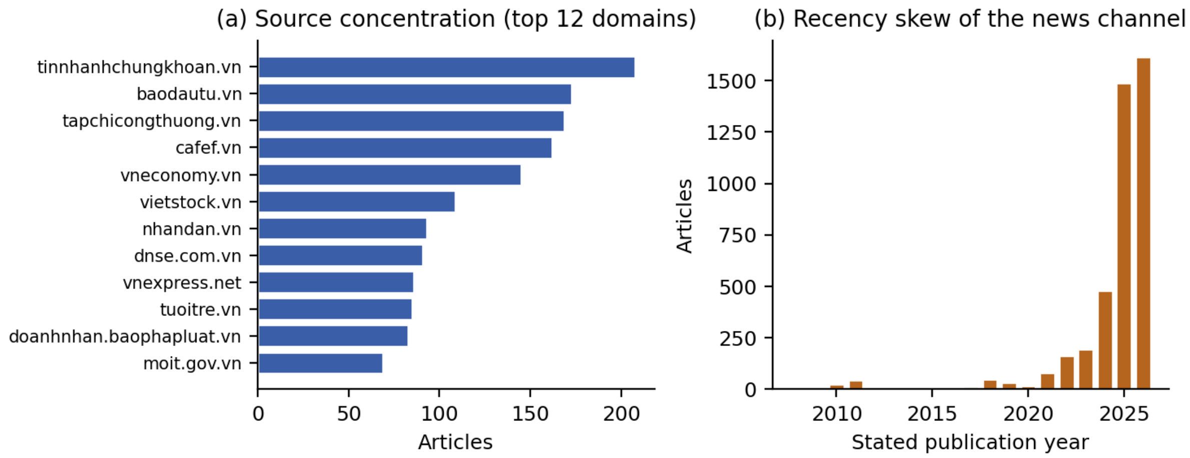

Figure 2.7: The conduct channel. (a) Source concentration across the twelve most frequent domains. (b) Distribution of stated publication year.

Table 2.6: Composition of the conduct channel, deduplicated across tickers; the same figures cited in the Abstract, Contribution 2 and Table 2.2.

Quantity	Value
Tickers covered	115
Articles retrieved	3,797
Sentences	174,256
Mean articles per ticker	33.0
Mean sentences per article	45.9
Distinct source domains	**716**
Share held by the three largest domains	12.1%
Retrieved via Google News / Bing / DuckDuckGo	2,844 / 931 / 22
Sentences explicitly mentioning the company	35.9%
Articles dated before 2010 (probable parse failures)	**220 (5.8%)**

Independence is structurally sound. Figure 2.7(a) shows the twelve most frequent domains: the 3,797 articles come from 716 distinct domains, and the three largest, `tinnhanhchungkhoan.vn`, baodautu.vn and tapchicongthuong.vn, account for only 12.1% of the corpus. This fragmented source distribution is the channel's most favourable property. No single outlet's editorial position can dominate the conduct side, though the corpus still skews toward financial and business media over environmental or local reporting, which bounds the kind of conduct it can observe.

Retrieval, however, is less balanced than the source list suggests. Of the three search back-ends in Table 2.6, one provider supplies roughly three-quarters of the corpus, far more than the three-back-end design (a hedge against single-provider ranking bias) was meant to allow. This is a limitation of the corpus as built, and it is recorded as such in §5.3.

A third issue shows up in Figure 2.7(b): the distribution of stated publication year is heavily skewed toward the present. 2,594 of 3,771 dated articles (68.8%) were published in 2025 or 2026, so the conduct channel is concentrated in the last two years while the claim channel spans fifteen. This time asymmetry, layered on top of the volume asymmetry documented in §2.3.6, is the direct cause of the asymmetric retrieval-window design in the Methodology chapter's cross-check stage. A claim made in 2012 has essentially no contemporaneous conduct evidence, so the window has to look forward, and far.

Finally, the pre-2010 dates flagged in Table 2.6 are not credible for a corpus of online financial news about currently-listed issuers. One value alone, 2002, accounts for 150 of the 220, which looks like a template default, not a real date. These are extraction failures, not old articles, and they are left in the corpus, uncorrected but flagged, since the date_uncertain contract of §2.2.5 exists precisely to carry this uncertainty forward to the reader instead of hiding it. This measurement is the empirical justification for that contract: without it, roughly one article in seventeen would contribute a fabricated year to a temporal graph.

2.3.6. Claim–conduct asymmetry

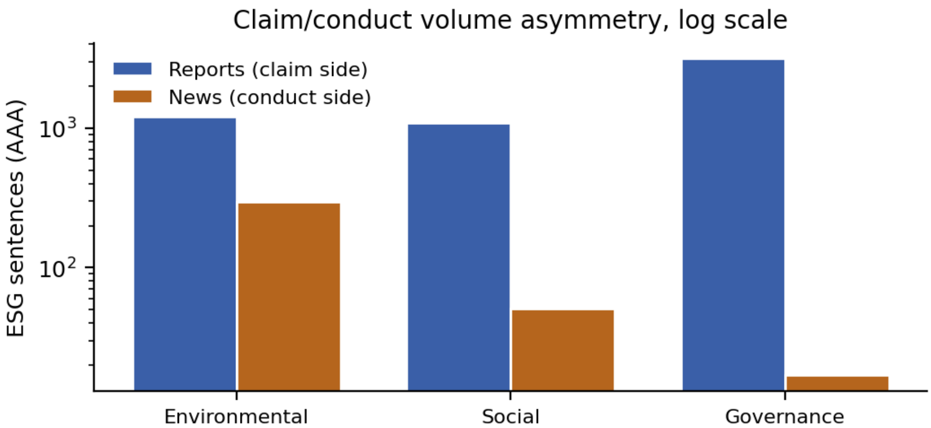

Figure 2.8: Volume and pillar asymmetry between the two channels for the issuer analysed end-to-end. Note the logarithmic vertical axis.

Table 2.7 and Figure 2.8 state the central empirical constraint on the entire system, and it conditions every cross-check result reported in the Experiment chapter: for the issuer analysed end to end, there is roughly one piece of independent conduct evidence for every sixteen ESG claims, before any consideration of whether the two are even about the same topic.

Table 2.7: Claim channel versus conduct channel for the issuer analysed end-to-end.

Quantity	Reports (claim)	News (conduct)	Ratio
Sentences	13,541	1,054	12.8×
ESG sentences	**5,575**	**352**	**15.8**×
ESG share	41.2%	33.4%	-
- Environmental	1,207	297	4.1×
- Social	1,084	51	21.3×
- Governance	3,157	17	185.7×

The pillar composition inverts as well. Governance is the largest pillar in reports and the smallest in news, while Environmental runs the other way (Table 2.7), so the governance ratio (186:1) is far more extreme than the environmental ratio (4.1:1).

The implication is fairly specific. Independent media write about factories, emissions and environmental incidents, not a company's internal control framework, so a governance claim ends up structurally far less checkable than an environmental one in this corpus. That isn't because governance claims are harder to evaluate in principle; it's because no independent party produces evidence about them. Since the claim side is governance-heavy and the conduct side is environmental-heavy, the two channels are least aligned exactly where claim volume is greatest. This is the structural reason for the large unverified count in the Experiment chapter's cross-check results. It is a property of the evidence environment, not a defect of the matching algorithm.

2.3.7. What the exploration implies for the design

Five properties measured above drive design decisions in Chapter 3, and they are collected here so the connection is explicit rather than implied.

1. The conduct channel is roughly 15.8× smaller than the claim channel and inverted in pillar composition (§2.3.6). The system therefore has to be able to return *"not verifiable"* as a first-class outcome instead of forcing a verdict, which is why the cross-check emits three assessments, one of which is `unverified_insufficient_evidence` (§3.1).

2. The conduct channel is concentrated in the last two years while claims span fifteen (§2.3.5). The retrieval window is deliberately asymmetric as a result, one year backward and fifty forward, a design specified in the Methodology chapter's cross-check stage.

3. 5.8% of articles carry an implausible publication date (§2.3.5). The `date_uncertain` flag is mandatory on news-derived observations because of this, and it is propagated into every dossier as a visible caveat (§2.2.5), a contract honoured throughout the cross-check stage described in the Methodology chapter.

4. The ESG classifier's 34.8% positive rate is an upper bound, with a demonstrated lexical failure mode (§2.3.3). The ESG label is used only as a cost filter over which pages to send to an expensive extractor, and it is never treated as evidence in its own right (§3.3).

5. 79.5% of extracted KPI observations fall outside the controlled vocabulary, and units are written 189 different ways, a KPI-yield result reported in the Experiment chapter. Canonicalisation is therefore a separate offline stage that writes a *new* property instead of overwriting the raw one, and it rejects rather than force-maps ambiguous cases, as described in the Methodology chapter's Validate & Normalize stage.

3. Methodology

3.1. Problem Definition

This section fixes the task the rest of the chapter operationalises, so supports, contradicts and unverified carry one meaning throughout, and so what the system does not do reads as a deliberate scope decision rather than an omission a reader has to notice on close reading.

Let O be the set of issuers (companies) in the corpus. A *sustainability claim* is a tuple $c = (o_c, \tau_c, \pi_c)$: $o_c \in O$ is the issuing organisation, τ_c is the interval over which the claim asserts something to be or become true (`valid_from`/`valid_to`, defaulting to the report's fiscal year when no explicit horizon is stated), and π_c is its provenance, the (document, page, sentence) triple it was extracted from. C is the set of all claims; every $c \in C$ carries `source_type = report`, since a claim is by definition something a company said about itself.

Let E be the set of *conduct evidence*: instances of the graph's news-derived observation classes (`KPIObservation`, `Controversy`, `Penalty`, `MediaReport`), each stamped `source_type = news` and each a tuple $e = (o_e, \tau_e, \pi_e)$ in the same shape as a claim. E is drawn from a corpus collected independently of the companies under study, the independent conduct channel introduced in Chapter 2 as this thesis's second contribution. This independence is what lets conduct evidence falsify a claim rather than merely restate it; the adjudication step below contributes nothing to that property on its own.

For a claim c, retrieval selects a candidate set $R(c) \subseteq E$ of conduct evidence sharing c's resolved issuer and temporally compatible with τ_c. This is made possible by the graph's entity resolution and temporal metadata, so the system never runs a full cross-product search over every claim and every article. An *adjudication function*

$$f : C \times E \to \{\mathsf{supports}, \mathsf{contradicts}, \mathsf{irrelevant}\}$$

is applied to every pair (c, e) with $e \in R(c)$, specified later in this chapter as a prompted LLM judgement over retrieved evidence, not a learned classifier. Claim-level status is then the aggregate

$$g(c) = \begin{cases} \mathsf{contradicted} & \text{if } \exists e \in R(c) : f(c,e) = \mathsf{contradicts} \\ \mathsf{supported} & \text{if no such } e \text{ exists, and } \exists e \in R(c) : f(c,e) = \mathsf{supports} \\ \mathsf{unverified} & \text{otherwise } (R(c) = \emptyset, \text{ or every } f(c,e) = \mathsf{irrelevant}) \end{cases}$$

Contradiction dominates support by construction: a single credible counter-example is informative even alongside corroborating evidence, while the absence of a counter-example proves nothing. Unverified deliberately collapses two situations a reader might want distinguished, that no candidate evidence existed for c, or that candidate evidence existed but was judged irrelevant, into one label, since both mean the same thing operationally: the pipeline has nothing that speaks against the claim, which is not the same as confirming it.

No function $h : C \to [0,1]$ is defined anywhere in this design. Composing g into such a score would have been technically straightforward, and it was deliberately left out: no ground-truth greenwashing label exists for this corpus, or for any Vietnamese corpus, against which h could be fit or validated. Returning $g(c)$ together with the evidence and provenance that produced it, and leaving the greenwashing judgement to the reader, is the scope boundary this thesis holds to.

3.2. System Architecture

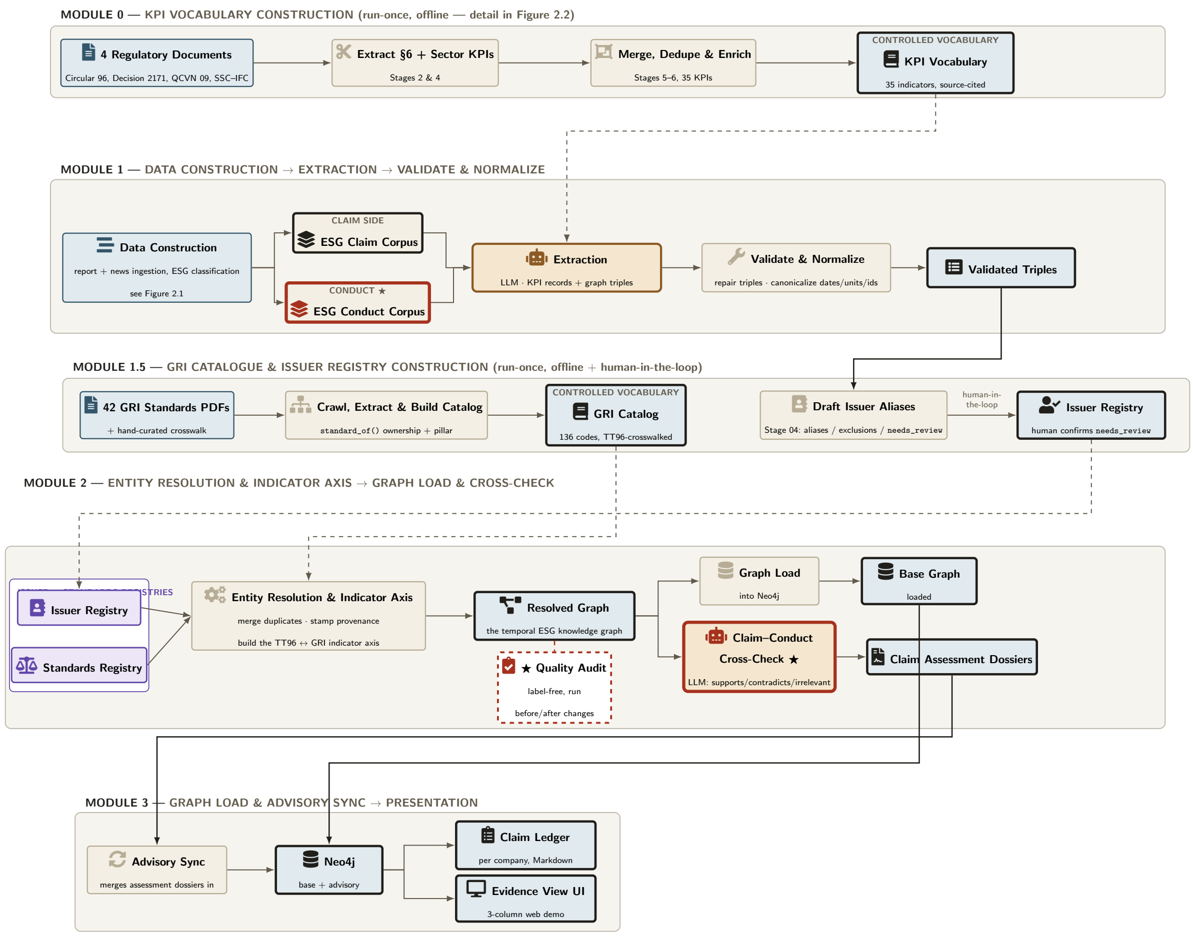

Figure 3.1: Full construction and analysis pipeline, redrawn as five stacked modules to fit the page width.

Figure 3.1 gives the complete system architecture referred to throughout this chapter. Data construction is detailed in full in Figure 2.1, so it is condensed here into a single module producing the two evidence corpora. The two reference-construction modules (0 and 1.5) show inline how the KPI vocabulary, GRI catalogue and issuer registry that later stages depend on are each built. What follows is the temporal-knowledge-graph construction and claim–conduct cross-check pipeline (Extraction, Validate & Normalize, Entity Resolution & Indicator Axis, Graph Load, Claim–Conduct Cross-Check, Advisory Sync, Presentation), which the rest of this chapter specifies one stage at a time. Blocks with a crimson border are this thesis's own contribution beyond the underlying EmeraldMind design; every other block is standard NLP or data-engineering machinery.

3.3. Extraction

Extraction is the stage that turns one page of ESG-labelled text into two structured artefacts: a set of KPI records checked against the controlled indicator vocabulary built in Chapter 2, and a set of typed graph nodes and directed, temporally-scoped edges checked against the class and edge-label schema from Chapter 1. Both come from a single language-model call per page, restricted to pages with at least one ESG-relevant sentence, and both preserve the (document, page, sentence) provenance of every value they emit. Identity resolution, cross-checking and the final claim ledger all depend on this provenance downstream, since none of them re-reads the source document.

The two calls are not independent. Figure 3.2 makes the dependency explicit: KPI extraction runs first, and its output is one of the inputs triple extraction receives, alongside the page text and the schema. A KPI record is more constrained than a free-form triple, since it can only take the shape one of the 35 controlled indicators defines, so resolving it first gives triple extraction a small, already-typed KPI observation to attach to an organisation, rather than asking one model call to invent both the indicator identity and the graph structure around it at once.

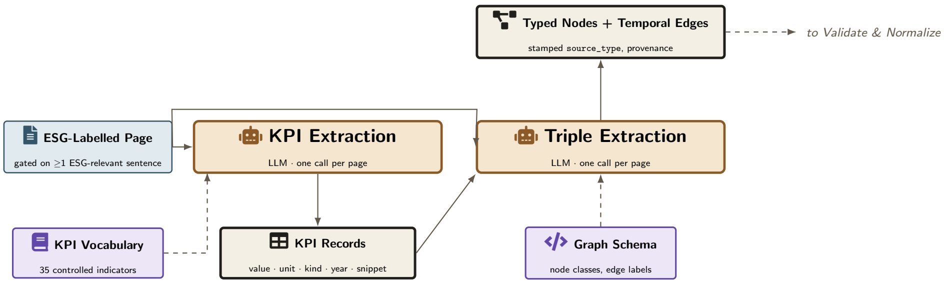

Figure 3.2: Extraction, detailed — internal structure of the *Extraction* block in Figure 3.1.

A single worked example fixes what these two calls actually produce. Page 31 of one 2024 annual report in the corpus follows a table of environmental-compliance figures with the sentence:

> *"Không có sai phạm về vấn đề môi trường xảy ra trong năm."*
> (No environmental violations occurred during the year.)

This sentence matches the controlled indicator for the number of environmental-compliance penalties incurred in the reporting period. Table 3.1 shows what each of the two calls emits for it.

Read in isolation, this is an entirely unremarkable, self-reported compliance record. It is also exactly the kind of record the independent conduct channel exists to check: a company has every incentive to report zero penalties whether or not one occurred, and nothing available to this stage (the page text, the indicator vocabulary, the schema) can tell an actually clean record apart from an unreported one. Extraction stays deliberately silent on that question and carries a self-reported zero forward as a claim like any other. It is the cross-check later in this chapter, matching it against evidence this stage never sees, that is positioned to distinguish the two.

Table 3.1: What KPI extraction and triple extraction each emit for the sentence above.

Field	Extracted value
Indicator	Number of times penalised for environmental non-compliance in the reporting period
Value / unit / kind	0 / times / achieved
Reporting year	2024
Snippet (verbatim anchor)	"Không có sai phạm về vấn đề môi trường xảy ra trong năm."
Node produced	one `KPIObservation` node carrying the fields above, plus a validity interval covering the 2024 reporting period
Edge produced	the reporting organisation → that observation, carrying its own temporal metadata (`valid_from`, `valid_to`, `recorded_at`) separate from either node's
Stamp on both node and edge	`source_type = report`

3.4. Validate & Normalize

Extraction is one language-model call per page, so nothing in it sees the graph as a whole: two pages can invent slightly different spellings of the same edge label, a temporal field can be left off, or the reversed direction of a legal (source class, target class) pair can slip through. Validate & Normalize closes that gap before any node is merged or resolved. It never touches the graph's identity or structure; its only job is to check that every extracted triple is schema-legal and that every date on it means the same thing everywhere else in the corpus. It runs three passes over the same triple set: a repair pass that corrects and temporally normalizes what extraction produced, an anchoring pass that adds structural links the extraction prompt could not always infer, and a canonicalization pass that places every KPI observation into the 35-indicator controlled vocabulary. The three run as a single unit that reads the triples once and writes the validated result once, rather than as three stages that each separately read and overwrite a shared intermediate file. This matters because the repair pass is the only paid step in the chain, and separating the passes would risk a later bug silently discarding work already paid for.

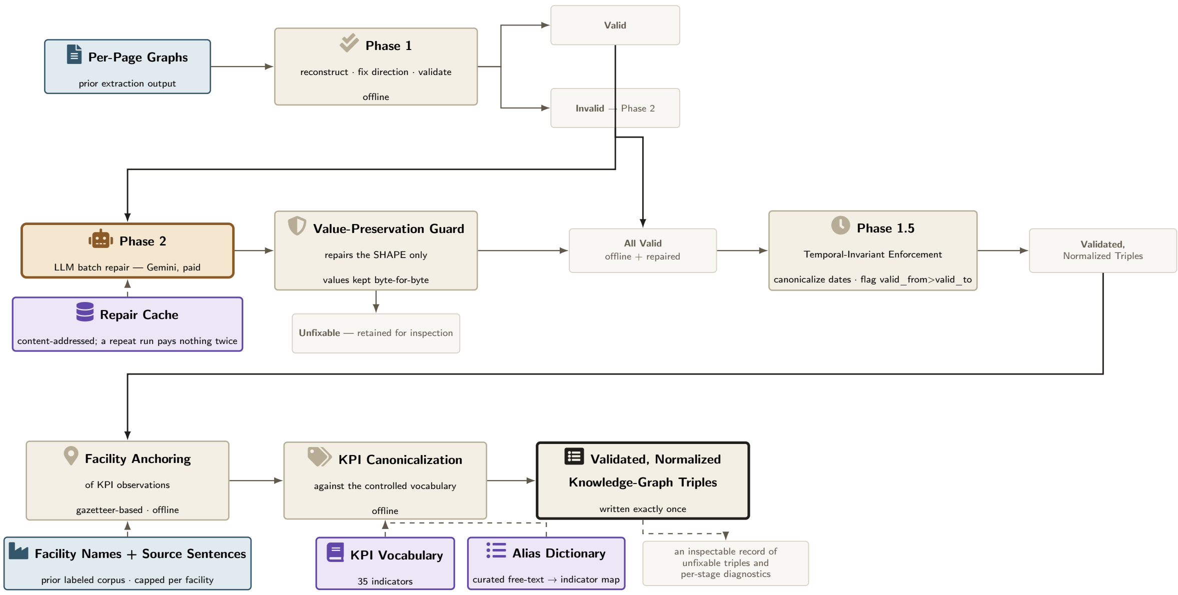

Figure 3.3: Validate & Normalize, detailed — internal structure of the *Validate & Normalize* block in Figure 3.1.

Every triple recovered from extraction is first checked against the graph schema, before anything else happens to it. A predicate's declared (source class, target class) pairs are consulted first: if a triple's subject and object appear only in reversed order, they are swapped rather than discarded, since a single edge label may legitimately run in more than one direction depending on which classes it connects (a claim's verification, for instance, may point at either a third-party verifier or a KPI observation). What remains is checked against the full schema contract from Chapter 1: every node's class must be a declared entity class, every edge's predicate a declared label, every node must carry a validity interval and a current-state flag, and every edge its own temporal metadata separate from either endpoint's.

Triples that fail this structural check are batched and sent to a large language model with one instruction this stage is built around: correct the shape of a triple (its class, predicate, or temporal fields), but never alter any other property value. That instruction is enforced in code, not only in the prompt. Every property the model returns is discarded and replaced with the corresponding value from the original triple, except the handful of fields it was explicitly asked to repair, and every attempted change outside that set is counted rather than silently applied. This guard is necessary rather than merely cautious, because schema validation alone cannot catch the failure mode it prevents: a repair that quietly translated a Vietnamese organisation's name into English would still pass every structural check, then split what should be one entity into two once identity resolution runs, since name matching treats a Vietnamese spelling and its English translation as different keys. Repairs are cached by the content of the triple, not its position in a batch, so a later re-run over the same invalid triples never pays for the same correction twice and reproduces a past failure exactly rather than asking the model again.

A worked example shows what the guard enforces in practice. On page 5 of AAA's 2013 annual report, triple extraction reads a line from an organisational chart and emits an edge from the reporting organisation to one of its departments using the predicate `hasManagementStructure`, which is not one of the 76 legal (label, source class, target class) combinations the schema declares (§1.3), so structural validation flags it and routes it to Phase 2. Table 3.2 shows what the batch repair returns for it, and what the guard keeps or discards.

Table 3.2: One invalid triple repaired through Phase 2, under the value-preservation guard.

Field	Before Phase 2	After Phase 2
Subject	Organization "AAA"	Organization "AAA" (unchanged)
Predicate	`hasManagementStructure` (no such edge label in the schema)	`owns` (a declared Organization → Organization edge)
Object	Organization "P. HC-TC"	Organization "P. HC-TC" (unchanged)
`valid_from` / `valid_to` / `is_current` (both nodes)	2013-01-01 / null / true	2013-01-01 / null / true (restored from the original, not taken from the model's reply)

Every date on every triple surviving the two passes above, node or edge, is canonicalized next to a single ISO representation, so two spellings of the same instant (a bare year and a full date) can never again be read downstream as two different facts. A date that cannot be parsed is counted rather than guessed at, and a fact whose start postdates its own end is flagged rather than silently corrected, since either boundary could be the one in error. A conduct-side observation with no explicit marker of date certainty defaults to the conservative reading: the date on record stands in for an unstated event date, not a claim that the event happened exactly then.

A further pass anchors KPI observations to the facilities they describe. A measured value with no link to the physical facility it was taken at is a fact floating free of the entity it describes, and extraction cannot always supply that link from a single sentence in isolation. This pass closes part of that gap offline, without a further model call. It builds an index of facility names already in the graph, locates each KPI observation's original source sentence, and adds the missing structural link only when that sentence names a facility already known to the graph. A cap on how many observations a single facility name may anchor guards against a common failure of this kind of matching: a name broad enough to match a large share of the corpus is a generic term, not a specific plant, and treating it as one would manufacture a false hub instead of a real anchor. No new relationship type is introduced; every link added is one the schema already permits.

The final pass canonicalizes KPI observations against the controlled vocabulary, giving every KPI observation a single, comparable code from the 35-indicator vocabulary (Chapter 2), instead of leaving it as whatever free-text wording the source report happened to use. A record already coded against one of the four regulatory sources is accepted as-is. For everything else, four matching rules are tried in order (exact alias, official indicator name, longest matching phrase, then a high-confidence fuzzy match), and the pass stops at the first one that fires, recording which rule matched so a wrong mapping can be traced back to it. The rules favor precision over recall: if none of them confidently fires, the observation is left unmapped rather than forced onto the nearest guess. Mapping never destroys information; the original wording is kept alongside the canonical code, not replaced by it. The same pass also does some light bookkeeping: normalizing units (rescaling to a common base unit when the conversion is known), deriving a reporting period, and backfilling a goal's target date from a year mentioned in its own text when that year is missing and later than the goal's start.

Validate & Normalize changes what a triple says about itself; deciding that two triples describe the same real-world entity is the next stage's task. Entity Resolution & Indicator Axis takes the triple set this stage produces, written once, after every correction above, and collapses duplicate mentions of the same organisation, facility and standard into single canonical nodes while keeping their temporal history intact.

3.5. Entity Resolution

A validated triple is internally consistent, but it does not yet know whether two triples describe the same real-world organisation, whether a claim can be traced back to a page a reader could open, or how a Vietnamese-vocabulary KPI compares to its GRI equivalent.

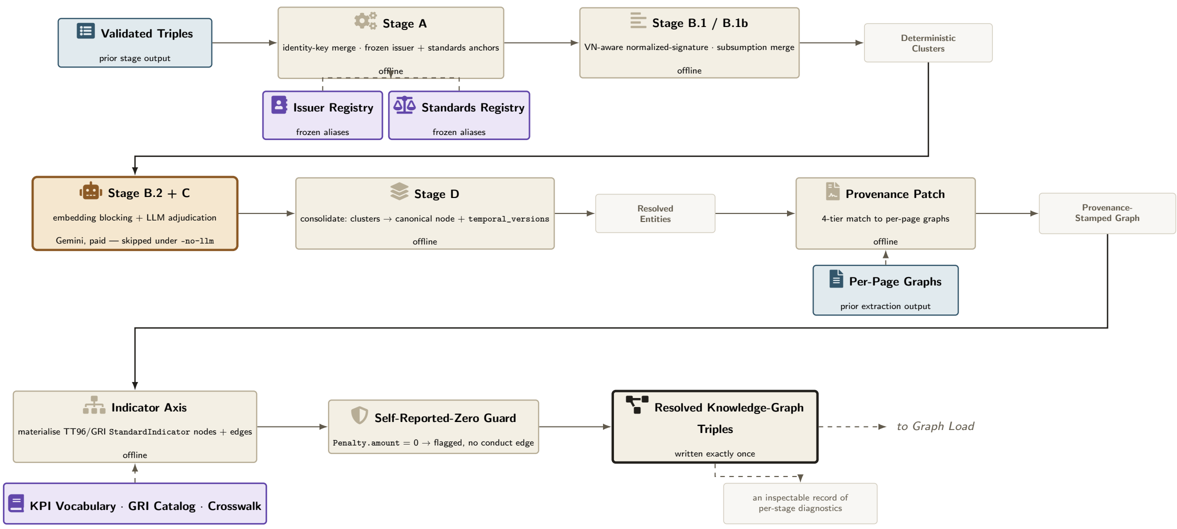

Figure 3.4: Entity Resolution & Indicator Axis, detailed — internal structure of the *Entity Resolution & Indicator Axis* block in Figure 3.1.

This first pass (Stage A in Figure 3.4) collapses only the duplicates it can be completely certain about, at no cost beyond the pass itself. A byte-identical match comes first: two nodes whose `identity_keys` (the timeless subset of a class's properties fixed in the schema, Chapter 1) are identical strings merge immediately. Since `Organization`'s only identity key is its name, this rule alone takes the corpus this chapter reports on from 13,770 raw graph nodes to 11,049. Two further merges then apply anchors built earlier in the pipeline and frozen, read here but never rewritten: the issuer registry (Figure 3.1, Module 1.5) supplies, per company, a human-confirmed list of every legal-name variant, alias and misspelling its reports are known to use, so a raw mention need not be textually close to its canonical form to be folded in; the standards registry does the same for GRI/TT96/QCVN document mentions. Freezing both anchors here, rather than re-deriving them at resolution time, is what lets a human correction made once survive every later re-run of this stage.

A worked example shows the issuer anchor collapsing a real spelling history. The organisation behind ticker ACC is mentioned 74 times across its annual reports in the corpus, spanning 2008 to 2024, under at least a dozen distinct spellings: legal-form variants, all-capitals scans, a short form without any legal suffix, and what the mentions' own dates suggest is a genuine trading-name change from the long-form legal name to *Becamex ACC* around 2024. None of those 74 raw strings are byte-identical, so the identity-key merge above leaves every one unmerged. The issuer anchor catches all 74, since `config/issuer_registry.json`'s `ACC` entry lists seven of these spellings as confirmed aliases and the rest normalise to one of the seven. Table 3.3 shows what the consolidation step keeps of the merge.

Table 3.3: One entity resolved through the frozen issuer anchor.

Field	Value
Raw mentions merged	74, dated 2008-06-01 to 2024-01-01
Sample of distinct spellings	"Becamex ACC"; "BECAMEX ACC"; "CÔNG TY CỔ PHẦN BÊ TÔNG BECAMEX ACC"; "Công ty Cổ phần Bê tông Becamex"; "ACC"; "Công ty Cổ phần ACC"
Matched via	Stage A.2, frozen issuer anchor (ticker `ACC`, 7 confirmed aliases in `config/issuer_registry.json`)
Canonical `name` (Stage D output)	"Công ty Cổ phần Bê tông Becamex" (the registry's legal name)
Current version (`is_current = true`)	"Becamex ACC", `valid_from = 2024-01-01` (the most recent trading name)
Other 73 mentions retained as	`temporal_versions` entries, each with its own `valid_from` and `is_current = false`

What survives the first pass is textually distinct but may still name the same entity under a spelling, capitalisation or diacritic difference the issuer registry does not list. The next merge instead compares a normalized signature (lowercased, diacritics stripped, corporate-form boilerplate such as "Công ty Cổ phần" or "TNHH" removed), so two spellings of one name collapse without any model call, and a further pass folds in a signature whose non-empty fields are merely a subset of another's, instead of requiring an exact match on every field. Only what remains ambiguous after these free passes is handed to an embedding-blocking step (cosine similarity under `gemini-embedding-001`) and an LLM adjudication step over the resulting candidate pairs (`gemini-2.5-flash`); these are the only two steps in this block that cost money. Both currently run under `-no-llm`: 6,261 raw entity mentions resolve to 3,202 canonical entities, a 48.9% reduction, from the free deterministic and Vietnamese-aware passes alone, with zero LLM comparisons made. A final consolidation step then folds each surviving cluster into one canonical node, keeping every distinct raw mention as a `temporal_versions` entry rather than discarding it, so the merge above is auditable by inspection even though it is not reversible in the data. Entity resolution's net effect on the graph's overall node count is a reduction from 13,770 to 10,598 (23.0%), observation nodes included.

A claim or observation node is only as useful as the page a reader can open to check it, and extraction (§3.3) does not always have enough context on a single page to self-stamp that link with certainty. This offline pass closes the gap: for every node in the schema's nine provenance-bearing classes it tries, in order, a parseable `source_id`, an exact `source_id` index lookup, a recomputed stable identifier, and finally a coarser page-token match against the per-page graphs extraction wrote, stopping at the first tier that succeeds and stamping `source_doc/source_page` (plus `article_title/url/domain` for a news document) from wherever it lands. It never reorders a node, since Graph Load keys Neo4j nodes by array position. In this run, every one of 7,287 provenance-bearing nodes, across eight of those nine classes (no `Controversy` node exists in this corpus), already carried `source_doc/source_page` at extraction time, so nothing was patched, and each was logged as `already_stamped` rather than skipped silently. The pass exists as a safety net for an extraction run that predates self-stamping, not because the current corpus needs it.

3.6. Indicator Axis

The last pass appends a shared axis on top of the now-identity-resolved, now-cited graph: a `StandardIndicator` node per regulatory indicator, a `measuredUnder` edge from a KPI observation to the indicator its `kpi_id` (§3.4) already names (never guessed here), an `equivalentTo` edge between a TT96 indicator and its GRI counterpart where the hand-confirmed crosswalk carries a confirmed row, and an `alignsWithIndicator` edge from a Claim, Goal or Initiative to the indicator its text discusses, resolved by a longest-matching-phrase keyword tier. This run adds 26 nodes (16 TT96/QCVN/SSCIFC indicators, 8 GRI indicators, 2 reference-document nodes) and 2,018 edges (1,300 `measuredUnder`, 649 `alignsWithIndicator`, 43 `partOf`, 26 `equivalentTo`) to the 10,598-node, 13,112-edge graph that entity resolution and provenance leave it. The pass is append-only, so the existing node/edge prefix stays untouched and a dossier computed against an earlier graph version stays valid. It also restamps every `StandardIndicator`'s pillar from the file entitled to say so (the Vietnamese vocabulary's own definitions, or the GRI catalog for a GRI indicator), 71 restamps here, never guessed from the indicator's own text, since the Evidence View UI reads that property directly for a claim's E/S/G column. One guard sits inside this pass rather than earlier: a `Penalty` node whose `amount` is 0 is flagged `self_reported_zero` and given no `measuredUnder` edge, the same self-reporting asymmetry already flagged for the compliance KPI in Table 3.1 (§3.3), now guarded here on the conduct-edge side too.

A worked example shows what the guard withholds. On page 65 of AAA's 2024 annual report, extraction (§3.3) reads the line *"Số lần bị xử phạt vi phạm: 0"* (number of times penalised: 0) and produces a `Penalty` node with `amount = 0`, exactly as it is entitled to. This pass adds the flag and stops there. Table 3.4 shows the node before and after.

Table 3.4: One self-reported `Penalty` flagged by the indicator axis rather than connected.

Field	Value
Source	AAA's 2024 annual report, page 65
Raw description (extraction)	"Số lần bị xử phạt vi phạm: 0"
Node produced (extraction)	one `Penalty` node, `amount = 0`
Flag added (this pass)	`self_reported_zero = true`
`measuredUnder` edge created	none, withheld specifically for this node
Why	a self-reported zero and an unreported violation read identically off the page; only independent evidence, matched later by the claim–conduct cross-check, can tell them apart
Corpus-wide, this run	1 of 11 `Penalty` nodes triggered the flag

Entity Resolution & Indicator Axis produces the single artifact every later stage reads: a graph with duplicate mentions merged, every claim and observation traceable to a page, and a regulatory indicator axis materialised across two standards. An optional fourth pass, not part of the block above, can resolve the small remainder of Claim/Goal/Initiative nodes the keyword tier leaves unmatched by asking an LLM a topic-classification question rather than a supports/contradicts judgement, but the pipeline is complete without it. What this block writes still lives only as a JSON file on disk; Graph Load is the next stage's task, loading it into Neo4j as a property graph so a query can traverse it instead of scanning it.

3.7. Graph Load

Entity Resolution & Indicator Axis leaves a single JSON file: a static array of nodes and edges, each edge pointing at its endpoints by array position rather than by anything a query language can address. Graph Load turns that array into the property graph every later stage runs against, unrolling the "Graph Load" block of Figure 3.1 (Module 2) into the point at which `resolved_graph.json` becomes the Base Graph that the claim–conduct cross-check, the claim ledger and the Evidence View UI all read from downstream (Module 3). No entity is merged and no new fact is inferred here; the stage's only obligation is that a Cypher query returns exactly what was already decided upstream, edge for edge and version for version. Figure 3.5 shows the two passes that guarantee this.

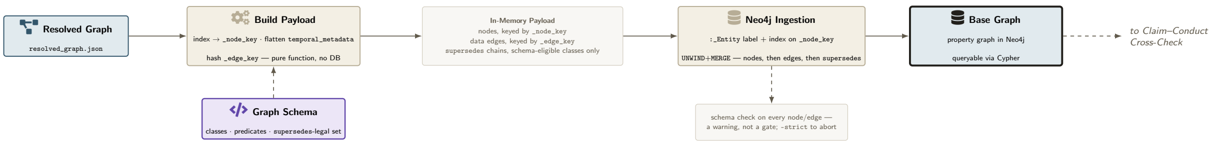

Figure 3.5: Graph Load, detailed — internal structure of the *Graph Load* block in Figure 3.1.

A node's identity in Neo4j is its position in the resolved array, stamped onto it as `_node_key = "n{i}"` during payload construction and never recomputed from the node's own properties. Every edge is rewired from the integer indices `resolved_graph.json` stores to those same keys before anything reaches the database, so the `MERGE` that creates a node keys on `_node_key` alone (a value Entity Resolution already fixed), not on any property a mis-typed diacritic or an OCR artefact could perturb.

Every edge's `temporal_metadata` is flattened onto the Neo4j relationship as three plain properties, `valid_from`, `valid_to`, `recorded_at`, and the relationship is keyed for `MERGE` on a deterministic `_edge_key`: the SHA-1 hash of both endpoint keys, the predicate, and all three temporal fields together. Keying on the endpoints and the predicate alone would collapse a fact that legitimately recurs across years into a single relationship on a second run, since `MERGE` treats an existing match as nothing further to do; an issuer located `isIn` the same province in 2017, 2020 and 2024 is three facts, not one restated three times. On the corpus this chapter reports on, one `isIn` relationship recurring across nine distinct years is written as nine separate relationships, each surviving ingestion under its own `_edge_key`.

`supersedes` is declared in the schema (Chapter 1) as a same-class self-edge, legal for only nine of the resolved graph's classes: Organization, Person, Facility, Goal, Product, Regulation, Standard, Material and Certification. For a node of one of those nine classes with more than one `temporal_versions` entry, this stage materialises the full history as a chain of its own nodes, newest first, headed by the canonical node (`is_current = true`) and linked `canonical`

`-[:supersedes]-> newest -> ... -> oldest`. For every other class, the version list is kept as a single JSON-string property on the one node instead, since minting a `supersedes` edge there would be schema-illegal. The choice is made per class by consulting the same schema every other stage validates against, not by a convention this stage invents on its own, so a run's Organization node with 74 raw mentions merged (Table 3.3) materialises as 74 chained version nodes, while a class outside the nine keeps its history as a single property instead.

Every node's class and every edge's predicate is checked against the schema a second time during ingestion, which is cheap insurance against a regression upstream rather than a repeat of validation already done; on the corpus this chapter reports on, that check raised no warnings. Because every write is a `MERGE` keyed on `_node_key` or `_edge_key`, re-running the loader against an unchanged `resolved_graph.json` changes nothing. This Base Graph is what the next stage queries: §3.8 retrieves, for every `SustainabilityClaim` this graph now holds, the conduct-side evidence sharing its resolved issuer and temporally compatible with it, and adjudicates each pair the way §3.1 already defined.

3.8. Claim–Conduct Cross-Check

This section is where the formal adjudication $f : C \times E \to \{\mathsf{supports}, \mathsf{contradicts}, \mathsf{irrelevant}\}$ and the claim-level aggregate $g(c)$ defined in §3.1 stop being notation and become code, the one stage where the claim side and conduct side are actually brought into contact. Consistent with this thesis's decision never to emit a greenwashing score (§1.3), the stage writes only these three categorical assessments, never a continuous score.

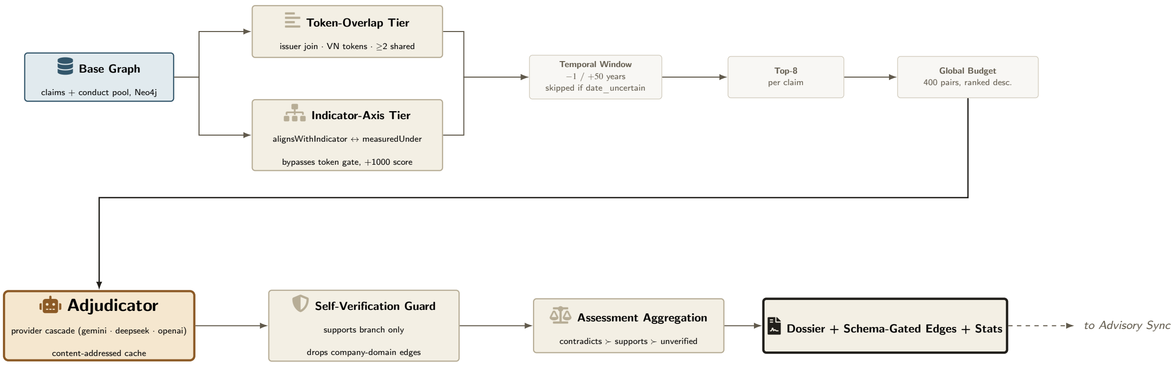

Figure 3.6: Claim–Conduct Cross-Check, detailed — internal structure of the *Claim–Conduct Cross-Check* block in Figure 3.1.

For a claim c belonging to issuer o_c, the candidate pool is first restricted to the graph's news-derived observation classes (`Controversy`, `Penalty`, `MediaReport`, `KPIObservation`, `ThirdPartyVerification`) stamped to the same issuer. Two independent tiers then decide which same-issuer candidates are worth sending to an LLM at all. The token-overlap tier Vietnamese-word-tokenises both claim and candidate, normalises and stopword-filters the result, and admits a candidate only if at least two whole tokens are shared. The indicator-axis tier instead joins a claim to conduct through the graph rather than through text, claim *alignsWithIndicator* `StandardIndicator` *measuredUnder* conduct, so a claim can reach evidence sharing no literal vocabulary with it at all; indicator-tier candidates bypass the token gate entirely and are scored with a fixed boost large enough to always outrank a plain token match once candidates are ranked.

Both tiers respect the same asymmetric temporal window: evidence may predate a claim's year by at most one year but postdate it by up to fifty, unless the evidence's own date is flagged uncertain, in which case the window is not enforced at all. The asymmetry follows directly from the conduct channel being far more recent than the claim side (§2.3.5); the window has to look forward, and far, rather than back. Per claim, surviving candidates are capped to the eight highest-scoring; across all claims in a run, the pooled candidate pairs are ranked once more and only a fixed budget, 400 pairs for the runs this chapter reports (raised from the code's 300-pair default), is actually sent to the LLM, so a run over an issuer with many claims degrades by adjudicating its strongest-signal pairs first rather than failing outright.

Each admitted pair is then judged independently by a large language model instructed to read exactly one claim and one piece of conduct evidence and return one of `supports`, `contradicts` or `irrelevant` with a confidence score and a rationale.

A `supports` verdict is not written back to the graph unconditionally, however. Before it becomes a `verifiedBy` edge, the evidence's source domain is checked against a curated list of the issuer's own web properties; a match drops the edge while keeping the evidence record itself visible in the dossier, marked not independent and carrying the reason it was dropped. This self-verification guard applies only on the `supports` branch: a company website corroborating its own claim is exactly the self-verification this thesis's conduct channel exists to rule out (§3.1), while a company website contradicting its own claim carries no comparable risk.

A claim's final status follows the priority the formal definition already fixed. If any admitted candidate was adjudicated `contradicts`, the claim is `appears_contradicted` unconditionally, regardless of how much independent support exists alongside it. Only in the absence of any contradiction does independently-sourced support, support that survived the self-verification guard, promote a claim to `appears_supported`. Everything else falls to `unverified_insufficient_evidence`, whether because no conduct pool existed, nothing retrieved was topically related, the run's budget could not reach every candidate, or every adjudicated candidate came back `irrelevant`. Every dossier also carries a standing caveat that the assessment is advisory and that no ground-truth greenwashing label exists to validate it against (§3.1), plus a further caveat whenever any evidence used has an uncertain date.

For every claim, one dossier records its assessment, its supporting and contradicting evidence (each with its retrieval tier, source, date, confidence and rationale), and its caveats. Graph edges are written alongside the dossier but only where the schema permits them: `verifiedBy` toward independently-supporting evidence, and `contradictedBy` or `contradictedByMedia` depending on the contradicting evidence's own class, so a verdict the schema has no legal place for stays dossier-only instead of being forced into the graph. Cross-Check leaves every result advisory and every edge it wrote gated by schema legality; it never asserts a graph-native fact on the strength of an LLM's opinion alone. Advisory Sync is the next stage's task: it takes exactly what this stage already paid to compute and folds it into Neo4j as a layer a reader can query, without spending another token.

3.9. Advisory Sync

Everything Claim–Conduct Cross-Check writes is still local: a dossier file per issuer and a JSON list of proposed edges, neither queryable until they reach Neo4j. Advisory Sync closes that gap with no further LLM inference, materialising the dossier Cross-Check already computed into the Base Graph §3.7 already built, as a layer a reader can tell apart from that base graph at a glance. The schema declares no legal edge from a `SustainabilityClaim` directly to a `KPIObservation`, so a claim a self-reported KPI number appears to contradict has nowhere to be written into the base graph itself; Advisory Sync's evidence edges are the one place that contradiction becomes a graph-native fact.

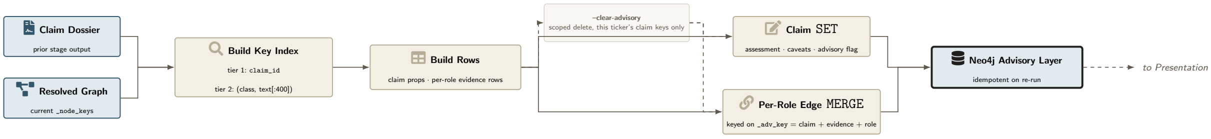

Figure 3.7: Advisory Sync, detailed — internal structure of the *Advisory Sync* block in Figure 3.1.

A dossier was written against whatever the resolved graph looked like when Cross-Check ran; by the time Advisory Sync runs, entity resolution may have re-run and shifted every array index. Trusting the dossier's recorded node index directly would silently bind an assessment to the wrong node after such a shift, so this stage instead resolves every claim and every evidence node against a fresh index built from the current graph.

The categorical `assessment` and its `caveats` list from §3.8 are then set directly on the existing `SustainabilityClaim` node, alongside a flag marking the whole block advisory, so a query can always tell what this stage concluded apart from what the base graph asserts.

Every piece of evidence in a dossier becomes one of three edge types depending on its role. Independently-sourced support becomes `llm_supports`. Contradicting evidence of any class becomes `llm_contradicts`, including, as above, a claim–`KPIObservation` contradiction the base schema has no legal edge for, so this is the one channel where that specific contradiction becomes a graph-native fact rather than a sentence buried in a dossier file. Support the self-verification guard already flagged as non-independent (§3.8) becomes its own distinct edge, `llm_flagged_support`, so a reader can still see what a company-owned source claimed without it ever being mistaken for independent corroboration.

Each edge is keyed for `MERGE` by the resolved claim key, the resolved evidence key and its role together, and every write is a scoped delete of this stage's own past output followed by a fresh `MERGE`, so re-running a sync against an unchanged dossier reproduces the same graph instead of accumulating duplicate edges.

What Advisory Sync writes is what the claim ledger and the Evidence View UI both read from downstream (Figure 3.1, Module 3); neither reruns any part of the pipeline above it, and both are presentation over exactly the graph this chapter has now finished building.

4. Experiments and Results

Chapter 3 specified what each stage does; this chapter reports what running them actually produced. The boundary fixed in §2.1.2 still holds: the KPI records, the validated triples and the resolved graph are outputs of the system under study, not corpus components, and they are measured here for the first time rather than assumed. No component of the pipeline is trained, so there is no accuracy metric computed against a held-out split; every number below is a yield, coverage or agreement statistic computed on the pipeline's own output, read directly from the stage-level statistics files each offline stage already writes as a side effect of running (§3.4, §3.5, §3.6). Two scopes are reported separately because the pipeline itself runs at two different scopes: §4.1 covers the full sector corpus Extraction and Validate & Normalize have processed to date; §4.2 covers Claim–Conduct Cross-Check, which is issuer-scoped by construction (§3.8) and has so far been run to completion for five issuers, AAA, ACC, ACG, ADP, AGG, rather than one.

Methodology's own worked example used a single issuer, AAA, to keep the mechanism legible; this chapter is where that illustration is widened into a result.

The two yield sections are followed by two evaluation sections, ordered by increasing evidential strength. §4.3 reports internal quality controls, which compare the system against its own design and are therefore a necessary condition rather than evidence of correctness. §4.4 reports the strongest result available: a blind annotation of the system's cited evidence by two independent domain experts, giving a precision figure and an agreement figure. Every quantity reported in those two sections is defined by an explicit formula before its value is given, and every proportion computed on a finite sample is accompanied by a 95% confidence interval. Table 4.1 fixes the notation those formulas use.

Table 4.1: Notation used throughout this chapter. The first block describes the graph and the cross-check output; the second describes the expert-annotation instrument of §4.4.

Symbol	Meaning
$G = (V, E)$	the resolved knowledge graph: node set V, directed labelled edge set E
$V_{T2\cup T3}$	event and observation nodes (`KPIObservation`, `Penalty`, ...)
Σ	the set of legal triples (predicate, source class, target class) declared by the schema
$\tau(e)$	the temporal metadata carried by edge e
D	the set of cross-check dossiers, one per `SustainabilityClaim`
$a(d)$	the advisory conclusion of dossier d, in {appears_supported, appears_contradicted, unverified}
P	the set of (claim, evidence) pairs the system cited as evidence
$S \subseteq P$	the cited pairs carrying expert labels
$\hat{r}(p)$	the relation the system asserts for pair p, in {supports, contradicts}
$h_k(p)$	the label expert k assigned to p, in {supports, contradicts, irrelevant}

4.1. Pipeline Yield: Extraction to Resolved Graph

Table 4.2: Validate & Normalize and KPI canonicalisation yield, read from the statistics the validation and canonicalisation stages emit as a side effect of running.

Quantity	Value
Triples kept (validated, including batch-repaired)	14,500
Triples dropped (unfixable after repair)	807 (5.3%)
Distinct `KPIObservation` nodes considered by canonicalisation	7,127
Mapped to the 35-indicator vocabulary	**1,464 (20.5%)**
Rejected / left unmapped	5,663 (79.5%)
– rejected for an out-of-scope unit (e.g. VND)	3,094 (54.6% of rejected)
Distinct raw unit spellings before normalisation	189

Table 4.2 shows only one KPI observation in five mapped onto the controlled 35-indicator vocabulary. Read alone, that looks like a low yield, but it is not simply an extraction failure. Canonicalisation tallies a mapping decision once per *distinct* `KPIObservation` node rather than per occurrence patched (7,324 occurrences map onto the 7,127 distinct nodes in the table, the denominator every percentage below is taken over), and its design rule favors precision over recall: an ambiguous case is rejected rather than force-mapped (§3.4). The single largest rejection reason, 3,094 of the 5,663 unmapped nodes (54.6% of all rejections), is exactly the case the rule targets: a value reported in a financial unit (VND, tỷ đồng) the regulatory indicator vocabulary has no legal place for, not a KPI the extractor misread. The other structural driver is unit inconsistency at the source: units are written 189 distinct raw ways before normalisation, an expected consequence of extracting from unedited natural-language text rather than a structured filing, which is why a rejected occurrence stays fully readable instead of being silently overwritten.

As Table 4.3 shows, the 23.0% node reduction (13,770 → 10,598) is the product of the complete deterministic, no-LLM Entity Resolution path, not of Stage A.1 alone. Stage A.1's identity-key equality rule, which merges exact-duplicate entities *and* observations since a fact patched twice under an identical signature is still one fact, takes the corpus from 13,770 to 11,049 nodes by itself, before either frozen anchor runs. Every stage after A.1 operates on the entity population alone: the 7,509 observation nodes are not candidates for merging under the frozen anchors, Stage B.1/B.1b, or Stage D. The issuer anchor folds 618 `Organization` mentions into 5 canonical clusters and the standards anchor applies the same frozen-registry mechanism to `Standard/Regulation` mentions; both are operation-level merge counts, not separate graph-size checkpoints, since no total-node snapshot was recorded between Stage A and Stage B.1. What *is* recorded is the entity population's own count: 6,261 raw entity nodes, none yet merged; 3,244 entity clusters once Stage A (A.1–A.3) and Stage B.1's offline VN-aware normalized-signature merge have all been applied cumulatively; and 3,202 final entity clusters after Stage B.1b's subsumption merge and Stage D's consolidation. `-no-llm` (recorded throughout this thesis, §3.5) only skips Stage B.2's embedding blocking and Stage C's LLM adjudication (`llm_comparisons`

Table 4.3: Entity Resolution and indicator-axis yield, read from the statistics the entity-resolution and indicator-axis stages emit as a side effect of running.

Quantity	Value
Raw graph, pre-resolution	13,770 nodes (6,261 entity / 7,509 observation)
Stage A.1 — identity-key merge (entities and observations)	11,049 nodes
– Stage A.2 frozen issuer anchor (entity-only merge count)	618 `Organization` mentions → 5 clusters
– Stage A.3 frozen standards anchor (entity-only merge count)	same mechanism, applied to `Standard`/`Regulation` mentions
Entity population, raw (before any entity-resolution stage)	6,261 nodes
– cumulative entity clusters after Stage A (A.1–A.3) + Stage B.1	3,244 clusters
– cumulative entity clusters after Stage B.1b + Stage D	3,202 clusters
Resolved graph (full block, no-LLM mode)	**10,598 nodes / 13,112 edges**
– node reduction	3,172 (23.0%)
Indicator axis added, this recorded run (§3.6)	+26 nodes / +2,018 edges
– `measuredUnder` created this run	1,300
– `alignsWithIndicator` created this run	649
– `partOf` / `equivalentTo` created this run	43 / 26
Final graph (persisted)	**10,624 nodes / 15,130 edges**
– `alignsWithIndicator` edges, cumulative on persisted graph	807
Claim/Goal/Initiative nodes w/ $\geq$1 cumulative `alignsWithIndicator` edge	718 / 1,421 (50.5%)
– `SustainabilityClaim` alone	299 / 481 (62.2%)

= `llm_matches` = 0 in this run); Stage A.1, both frozen anchors, and Stage B.1/B.1b all run regardless, and together with Stage D they are what the 23.0% figure reflects.

The indicator axis reaches just over half of every claim-like node in the persisted graph: 718 of the 1,421 `SustainabilityClaim/Goal/Initiative` nodes (50.5%) carry at least one `alignsWithIndicator` edge, and 299 of 481 `SustainabilityClaim` nodes alone (62.2%). The narrower, higher figure is expected, since a `SustainabilityClaim` more often names a concrete, indicator-shaped topic than a `Goal` or `Initiative` does. This is a structural property of the graph as persisted, available in principle to a graph-native join; whether the retrieval-side indicator-axis tier in §4.2 actually draws on it in a given run is a separate, empirical question, taken up there rather than assumed here.

The 649, 718 and 807 figures in Table 4.3 need not reconcile 1:1. `alignsWithIndicator` edges accumulate across runs of this append-only pass instead of being rebuilt from a clean graph each time (§3.6), so the three figures answer different questions: 649 is what this specific recorded run *created*; 807 is the *cumulative* edge count on the persisted graph, including edges written by earlier runs still present in the file; and 718 is how many distinct claim-like *nodes* those cumulative edges reach, which need not divide evenly into 807 since one node can carry more than one aligned edge. All three are individually faithful to their own source, drawn from different points in the graph's accumulation history rather than one snapshot.

4.2. Claim–Conduct Cross-Check Results

§2.3.6 already fixed the constraint every result below is read against: for the issuer analysed end to end in Chapter 3, there is roughly one piece of independent conduct evidence for every sixteen ESG-labelled sentences on the claim side. This section reports what Claim–Conduct Cross-Check actually returned once that constraint met real retrieval and adjudication, for every issuer the stage has been run against to date.

Table 4.4: Claim–Conduct Cross-Check, all five issuers with an end-to-end resolved graph. Read from the per-issuer statistics the stage emits; every run used the same adjudication provider, so the five columns are directly comparable to each other.

Quantity	AAA	ACC	ACG	ADP	AGG	Total
Claims	36	14	301	69	44	464
Conduct pool	68	52	190	3	29	342
Claims w/ candidate (%)	11 (31)	2 (14)	134 (45)	7 (10)	7 (16)	161 (35)
Pairs adjudicated	23	2	359	9	8	401
Signal: supported / contra.	0 / 0	0 / 0	12 / 3	0 / 0	1 / 0	13 / 3
Unverified (%)	36 (100)	14 (100)	286 (95)	69 (100)	43 (98)	448 (97)

The asymmetry's signature holds at every issuer in Table 4.4: 448 of 464 claims (96.6%) land on *unverified*, and three of the five issuers, AAA, ACC, ADP, return *no* supported or contradicted assessment at all. This is the same structural result §2.3.6 and §3.8 already predict, now shown to hold across issuers rather than resting on one. A conduct pool sixteen times smaller than the claim side, concentrated in the last two years while claims span fifteen (§2.3.5), leaves most claims with nothing topically related to check them against, and the system reports that absence as `unverified_insufficient_evidence` rather than forcing a verdict, the first-class outcome §2.3.7 names as the design's direct response to the asymmetry. Sixteen schema-legal `verifiedBy`/`contradictedBy` edges are written back into the graph from this run: fifteen for ACG, one for AGG (§3.9).

ACG is the exception, and the same table shows why. Fifteen of the sixteen `appears_supported` /`appears_contradicted` assessments in this run belong to ACG, whose conduct pool (190) is both the largest of the five in absolute terms and, at 1.19 candidates retrieved per claim, the densest relative to its own claim count, nearly double AAA's 0.64. Signal, then, tracks conduct-pool depth and density rather than claim volume: AGG and ADP have claim counts of the same order as AAA's, but conduct pools of only 29 and 3 respectively, and together account for only the sixteenth assessment (AGG's single supported claim, with ADP returning none). Where the independent evidence channel is thin, as it is everywhere in this corpus (§2.3.6), it is exactly the issuer with the deepest conduct pool that this design is positioned to say something about.

Table 4.5: Two real ACG assessments from the same 2022 dividend event, read in opposite directions by the same independent evidence.

Claim	Text (Vietnamese, verbatim)	Assessment	Independent evidence
`claim_e81c...`	"Nghị quyết của ĐHĐCĐ số 09-2022/NQ-GAC… thông qua phương án chia cổ tức năm 2021, Kế hoạch chi trả cổ tức năm 2022."	`appears_supported`	cafef.vn: a 50% stock dividend, 43.82M shares, 438 tỷ đồng face value, the resolution's own numbers, corroborated
`claim_1a45...`	"Công ty không trả cổ tức bằng phương thức 'Script dividend'."	`appears_contradicted`	The same 50% stock-dividend issuance (cafef.vn KPI observation and MediaReport) read by the adjudicator as exactly the payment method the claim denies using

Table 4.5 is worth reading as a pair rather than as two isolated rows. The same underlying event, a 50% stock dividend ACG's own AGM resolution describes in one part of the report, confirmed against the running Neo4j instance where both conduct nodes carry both an `llm_supports` *and* an `llm_contradicts` edge, independently supports one claim and, elsewhere in the same report, contradicts another that denies using that payment method by its English accounting name. The system did not need to be told these two claims were related; retrieval found the same conduct evidence for both from token overlap alone, and adjudication read the relationship in each pair independently and reached opposite, evidence-consistent verdicts. Whether "Script dividend" names a distinct instrument from the stock dividend the resolution describes, or the second claim is simply imprecise, is a question this thesis's advisory framing (§3.1) deliberately leaves to a human reader with the two source pages open; the system's contribution is surfacing the tension and citing exactly what it is built from, not adjudicating it further. Every assessment in both tables carries the same standing caveat this thesis has applied throughout: advisory only, against no ground-truth label (§3.1).

4.3. Internal Quality Controls

Every control below but one compares the system against *its own design*, not against the world (a schema-compliance rate, for instance, is computed by the same schema the validator enforces), so none of them can fail in an interesting way; treat them as a necessary rather than sufficient condition, and read §4.4 for the external evidence.

Briefly, what each measures: $M_{2.1}$ (temporal completeness) is the share of the graph's non-entity population, edges E plus event/observation nodes $V_{T2 \cup T3}$, carrying a usable temporal record (a complete $\tau(e)$ on an edge, a start date on a node; an open valid_to counts as present, not missing, so the metric never rewards inventing an end date). $M_{2.2}$ (schema compliance) is the share of edges whose (predicate, source class, target class) triple is schema-legal (Σ). $M_{5.1}$ (abstention rate) is the share of dossiers $d \in D$ resolving to $a(d) = \mathsf{unverified}$.

One control is not self-referential, and it is the strongest positive result in this section. Round-trip grounding returns every extracted KPI value to the exact report page its own provenance cites and checks that the number appears there verbatim, against the source document, which no pipeline stage can edit, so unlike the rest, this metric can genuinely fail. It evaluates to $3{,}912/4{,}001 = 97.78\%$: fewer than one extracted value in forty cannot be found on the page it claims to come from.

Table 4.6: Internal quality controls over the resolved graph and the cross-check output. Two rows are not measurable: the full-corpus sentence-level artefact they need was not present when this evaluation ran. $M_{4.1}$'s numerator (718) is the cumulative count from Table 4.3 (§4.1), not this run's own 649.

Code	Quantity	Value	Numerator / denominator
$M_{1.1n}$	ESG signal-to-noise, news channel	62.14%	47,990 / 77,229
$M_{1.2n}$	Source provenance rate, news channel	100.00%	174,256 / 174,256
$M_{1.1r}$	ESG signal-to-noise, report channel	*not measurable*	
$M_{1.2r}$	Source provenance rate, report channel	*not measurable*	
$M_{2.1}$	Temporal metadata completeness	93.02%	21,620 / 23,243
$M_{2.2}$	Schema compliance rate	100.00%	15,130 / 15,130
$M_{2.3}$	Value-preservation guard	100.00%	500 / 500
$M_{3.1}$	Timeless-identity violation rate	0.00%	0 / 14
$M_{3.2}$	Cluster conciseness	0.47%	10 / 2,135
$M_{4.1}$	Standard-indicator alignment coverage	50.53%	718 / 1,421
$M_{4.2}$	Zero-report self-praise exclusion	100.00%	1 / 1
$M_{5.1}$	Evidence asymmetry / abstention rate	96.55%	448 / 464
—	**Round-trip KPI grounding**	**97.78%**	**3,912 / 4,001**

Of Table 4.6's controls, only $M_{2.1}$ misses its target: 93.02% against 100%, meaning roughly one edge or event node in fourteen carries no usable start date and is invisible to any time-bounded query. Since every greenwashing question is time-bounded, a 2021 commitment read against 2023 conduct, this is the section's main structural limitation, not a cosmetic shortfall.

The final control is the only one built so the system can lose. Rather than checking the system against its own design, it tests the null hypothesis that cited evidence for issuer T is no more likely to come from T's own crawled feed than that feed's overall share of the conduct pool would predict (lift ≈ 1); a lift of 2 or more is read as genuine company-specific attribution. On the cited output measured here it passes outright: all 24 cited pairs draw solely on the examined issuer's own feed, zero cross-feed citations, though that 100% rests on only 24 items, giving a Wilson lower bound of 86.2%, a statement about attribution discipline on a narrow slice of the 464 claims, not about coverage. (The cited population has since grown to 43 pairs, §4.4; the control has not been re-run at that scale.)

This is also the thesis's one controlled before/after. The identical check, run on the 226-pair population that predated a 2026-08-07 adjudication-cache fix (§3.8), had failed outright at 28.76% (65/226), no better than base rate. But the fix bundled more than the cache key: it also issuer-scoped conduct retrieval, added a topic-overlap gate, and tightened the adjudication prompt, and the pre-fix run itself was adjudicated by OpenAI, a provider removed before the fix landed. The comparison is therefore diagnostic, showing that a real regression was caught and repaired, rather than a fully isolated, single-variable ablation.

Together, these controls establish structural, provenance and retrieval sanity, that the graph is well-formed and its citations are genuine, while §4.4 supplies the external, semantic evaluation of whether the verdicts built on top of them are actually correct.

4.4. Expert Annotation: Precision and Agreement

No labelled greenwashing dataset exists for Vietnamese listed companies, so the system cannot report precision against a gold standard. It can report precision against a blind, protocol-governed annotation of its own cited output by domain experts, which is standard practice in fields where no gold labels exist. This section reports that measurement.

4.4.1. The annotators

Two senior industry practitioners, referred to throughout this section as Expert 1 and Expert 2 to preserve their anonymity, labelled the cited pairs independently of one another. Both work in the sector the corpus covers, and both are external to the research team: Expert 1 runs a construction group in the same sector as the corpus and reads and signs off on the class of disclosure this system evaluates; Expert 2 assesses corporate borrowers in this sector professionally and routinely weighs a company's stated position against independent evidence. Neither took part in designing the system or the annotation instrument.

One clarification matters for how this result is read. The evaluation design of this project also contains a separate, more elaborate instrument: a five-point Likert rubric across four dimensions, scored by three distinct panels (chief executive, human-resources director, auditor) with expertise-weighted consensus. That study has not been carried out, and the annotation reported here is not it. The instrument used here is a three-way relation-labelling task, and the two experts above are annotators of that task, not a rubric panel. The distinction is preserved deliberately, since conflating them would overstate what was done.

4.4.2. Instrument and scope

The unit of analysis is a (claim, evidence) pair the system cited. Cross-Check writes a pair into a dossier only when the adjudicator judged it supports or contradicts; a pair judged irrelevant is discarded and leaves no trace (§3.8).

The stage's cited output currently comprises 43 (claim, evidence) pairs across the five issuers (37 supports + 6 contradicts), spanning 27 distinct claims and two issuers (ACG, 42 pairs; AGG, 1 pair). This was cross-checked directly against the Neo4j advisory layer the system serves from: every one of the 43 pairs matches an `llm_supports`/`llm_contradicts` edge one for one, with the same asserted relation, so the population labelled below is exactly what the running system currently cites, not a stale or offline copy of it. Every one of the 43 pairs was blind-labelled by both experts; this is a census of the stage's complete current cited output, not a sample of it.

Three properties of the protocol bear on how much weight the numbers carry. It was fixed in writing before any result was seen, so changing a criterion after seeing results would have voided the measurement. Blindness is enforced mechanically rather than by instruction: each row is assembled from an explicit whitelist of six display fields, so no field carrying the system's verdict, the evidence's originating company, the model's confidence or its rationale can reach an annotator even if such a field is added upstream later. And the label definitions bias against the system: an annotator torn between supports and irrelevant is instructed to choose irrelevant, which makes every precision figure below a lower bound rather than an upper one.

4.4.3. Precision

Let S be the set of cited pairs carrying expert labels, $|S| = 43$, the stage's entire current cited output. Precision for expert k is the fraction of them whose asserted relation the expert independently confirms:

$$\text{Precision}_k = \frac{\big|\{p \in S : h_k(p) = \hat{r}(p)\}\big|}{|S|}$$

Because $\hat{r}(p)$ is never irrelevant (a pair judged irrelevant by the adjudicator is never written to a dossier, so it can never enter P), this instrument measures precision only. Recall is not merely unmeasured but unmeasurable from it: evidence the system discarded leaves no trace to sample from, so any recall figure derived from it would be an overclaim.

Proportions are reported with Wilson score intervals [17] rather than the normal approximation, which at this sample size would extend beyond the unit interval:

$$\text{CI}_{95\%} = \frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \qquad z = 1.96$$

Table 4.7: Adjudicator precision against each expert's blind labels, over the 43 cited pairs carrying labels. Wilson 95% intervals.

Quantity	Expert 1	Expert 2
Precision, labelled cited pairs	**62.8%** (27/43) [47.9, 75.6]	**65.1%** (28/43) [50.2, 77.6]
– pairs asserted supports	59.5% (22/37) [43.5, 73.7]	62.2% (23/37) [46.1, 75.9]
– pairs asserted contradicts	83.3% (5/6) [43.6, 97.0]	83.3% (5/6) [43.6, 97.0]

Table 4.7 gives the result: both experts confirm a majority of the 43, twenty-seven and twenty-eight of them respectively, and their errors land in the same place. Table 4.8 gives the full distribution: every supports error for both experts is a judgment of irrelevant, never a reversal to contradicts, so on that branch the system is citing evidence with no real bearing on the claim rather than asserting the wrong direction. The one error on the contradicts branch is different in kind: both experts independently reverse the same pair to supports (a claim about flexible, risk-aware management, cited against a KPI reporting 93.1% completion of the 2022 revenue plan), the one case in this census where the system's asserted direction, not just its relevance judgment, is contradicted by both annotators.

Table 4.8: System-asserted relation (rows) against expert label (columns), over the 43 labelled pairs, one sub-table per expert.

	Expert 1			Expert 2		
System asserts	sup.	con.	irr.	sup.	con.	irr.
supports ($n = 37$)	**22**	0	15	**23**	0	14
contradicts ($n = 6$)	1	**5**	0	1	**5**	0

4.4.4. Inter-annotator agreement

Precision figures from a single annotator carry no information about whether the task is well posed. Agreement between two independent annotators does. On these 43 pairs the two experts assigned identical labels on 42 of 43 (97.7% raw agreement). The sole disagreement is a different pair from the one discussed above: a system-asserted supports pair, carried over from an earlier annotation round, where one expert reconsiders it as irrelevant on this pass (a claim about continuous improvement, cited against a news article on a Net Zero pledge) while the other keeps the supports label given previously.

Because observed disagreement is no longer zero, the chance-corrected coefficients are informative rather than trivial here: Cohen's $\kappa = 0.960$ [18] and Gwet's AC1 $= 0.967$ [19] agree closely with one another and both fall in the "almost perfect" band [20]. What the result supports is narrow but real: on the pairs this system actually cites, two senior practitioners working blind and independently agree with each other about what the evidence shows in all but one case, so the residual shortfall in precision (Table 4.7) is a property of the system's citations rather than of an ambiguous annotation task.

4.4.5. What this measurement can and cannot support

Four limits belong here rather than in a closing limitations section, because a reader who takes the headline figure without them will overstate it.

- The figure rests on forty-three pairs, six of them on the `contradicts` branch. The overall Wilson intervals span roughly 48 to 78 percentage points, and the `contradicts` interval alone spans 43.6 to 97.0 on $n = 6$. The defensible claim is that roughly two-thirds of what the system currently cites is confirmed by a domain expert, not a single point estimate read to three significant figures.

- The company-attribution check is incomplete for this population. Both experts recorded whether the evidence concerns the correct issuer only for the 19 pairs carried over from an earlier annotation round, leaving it blank on the 24 pairs newly labelled here. A same-company-restricted precision estimate, the natural complement to Table 4.7, is therefore not available for the stage's complete current output.

- The labelled pairs are concentrated in one issuer. 42 of the 43 pairs belong to ACG and the remaining one to AGG, spanning 27 distinct claims. The measurement is predominantly about one issuer's dossiers, and generalising it across issuers is not supported by this sample.

- Precision is measured on a narrow slice of the system's output. The stage cites 43 pairs across 464 claims (§4.2), so high precision on this output is a statement about what the system chose to show, not about how much of the available evidence it finds; §4.4 above establishes that this instrument cannot measure recall at all.

5. Discussion and Threats to Validity

Chapter 4 reported what the pipeline produced; this chapter reads those results together, states plainly what the strongest one can and cannot support, and names the threats a reader should weigh before trusting any of it beyond this corpus.

5.1. Reading the Results Together

Three findings from Chapter 4 are strongest read jointly rather than in the sections that reported them separately. First, the internal controls (§4.3) show a system that is structurally sound: 100% schema compliance, 97.78% round-trip grounding, a negative control that correctly failed before a real defect was fixed and correctly passed after. But structural soundness is a necessary condition, not evidence that any individual verdict is correct. Second, the external result (§4.4) is the only evidence in this thesis that does not merely check the system against its own design: two independent domain experts, working blind, confirm roughly two-thirds of what the system currently cites and agree with each other on all but one of 43 pairs. Third, both of those results are computed on a small, retrieval-limited slice of the system's total output, not on its full claim population. The reconciliation of that fact is the subject of §5.2 below, because omitting it is precisely the failure mode a reader of this thesis should be guarded against.

5.2. The Aggregate Evaluation Base

Every component of the number below is disclosed somewhere in Chapter 4; this section is the one place they are added up. Of the 464 claims cross-checked across all five issuers, only 16 (3.4%) ever receive a `supported` or `contradicted` verdict at all (§4.2); the remaining 96.55% abstain as `unverified_insufficient_evidence`, which the design treats as a first-class, honest outcome rather than a failure to force. Fifteen of those 16 verdicts belong to ACG, the remaining one to AGG; three of the five issuers evaluated, AAA, ACC, ADP, return no supported or contradicted assessment whatsoever. This 16-verdict count is the frozen evaluation-baseline snapshot (§4.2); the 43 (claim, evidence) pairs the two external experts blind-labelled (§4.4) are a later, separately-frozen census, taken after the adjudication-cache fix grew the cited population further, and span 27 distinct claims (42 pairs from ACG, 1 from AGG) rather than 16. The two counts describe the same growing population at different points, not a contradiction. The strongest empirical result this thesis reports is therefore a measurement of one issuer's dossiers, not a general property of the system across the five it was run against, and should be read as such.

A second aggregate belongs beside the first. The dual-standard indicator axis (§3.6) is the one retrieval path in this design that is genuinely graph-native: a claim reaches conduct evidence through a 2-hop structural join, claim `-alignsWithIndicator->` Indicator `<-measuredUnder-`

conduct evidence, rather than through lexical overlap. Across all 401 (claim, evidence) pairs adjudicated for all five issuers, that path contributed **zero** of the pairs the system ultimately cited; every cited pair, in every issuer, carries `retrieval_tier = token_overlap`. The cause is structural rather than a bug: news-side KPI observations are overwhelmingly reported in a financial unit (VND), and canonicalisation deliberately rejects an out-of-scope unit rather than force-mapping it (§4.1), so a conduct-side observation almost never acquires the `measuredUnder` edge the join needs, however well the claim side is aligned. The indicator axis therefore contributes schema, identity resolution, temporality and provenance to this system, all real, and all measured in Chapter 4, but not retrieval, in the run reported here. Retrieval in the reported system is Vietnamese token overlap with a temporal window; any reading of this thesis that treats claim–conduct matching as graph-structured retrieval overstates what actually ran, and should be corrected against this paragraph.

5.3. Threats to Validity

LLM calls in this pipeline are non-deterministic in a way no control here fully closes. Every prompted stage (extraction, repair, entity adjudication, claim–conduct adjudication) runs at `temperature=0`, and prompts are pinned byte-for-byte and guarded by tests specifically so a reworded prompt cannot silently change every downstream verdict. Neither control eliminates non-determinism outright: a hosted API is not contractually guaranteed to return a bit-identical completion for an identical request, no seed parameter is exposed to pin generation further, and the model identifier behind an API name can change server-side between runs without a corresponding change on this project's side. A re-run of this pipeline against the same input corpus is therefore expected to reproduce the same *shape* of output, schema-compliant, similarly distributed across supports/contradicts/irrelevant, but individual verdicts are not guaranteed to be bit-identical, and this thesis does not measure how much they vary.

The triples a claim and its candidate conduct evidence are built from, and the verdict that compares them, are both produced by prompted language models drawn from the same small set of provider families (Gemini, DeepSeek); an LLM, in other words, adjudicates evidence an LLM also extracted, and this is not named as a threat anywhere else in this thesis. A shared failure mode, a systematic misreading of a Vietnamese idiom, a consistent blind spot in how a particular KPI phrasing is parsed, could in principle look self-consistent to an adjudicator built the same way as the extractor, in a way it would not to a genuinely independent reader. The blind expert annotation of §4.4 is the direct mitigation for this specific risk, which is exactly why it is treated as the strongest result in this thesis rather than the internal controls; but that annotation covers 43 pairs concentrated in one issuer (§5.2), so the mitigation's own coverage is as narrow as the risk is broad.

Three threats are stated in full elsewhere in this thesis and are recorded here only so this list is complete. The evaluation base is narrow on both of its strongest legs, and the graph-native indicator-axis retrieval path fired zero times out of 401 adjudicated pairs; §5.2 states both precisely and gives the structural cause of the second, and neither should be read off this list alone. Separately, the resolved graph accumulates indicator-axis edges across runs rather than being rebuilt identically, since an append-only pass protects previously-paid cross-check dossiers from being invalidated by a later re-run, at the cost of the graph not being exactly reproducible from a clean rebuild (§3.6); §4.1 reconciles the resulting 649-versus-718-versus-807 figures in

Table 4.3. Taken together, they mean no claim in this thesis about precision or agreement should be read as established across the five issuers evaluated, still less across the sector the underlying corpus covers.

The annotation population also changed size during this project, and that history is not restated in Chapter 4. An earlier annotation round labelled 200 real pairs (plus 20 attention-check decoys) sampled from a 226-pair population, and found 26.5%/35.0% overall precision by the two annotators, with precision as low as 7.0%/8.8% on the `contradicts` branch specifically. That round predated a fix to the adjudication cache (§3.8) whose key had omitted the prompt, so a re-run had been silently replaying pre-fix verdicts; fixing it and re-running collapsed the cited population from 226 to 24 pairs, and a subsequent full re-run against the corrected cache grew it again to the 43-pair census Chapter 4 reports. By deliberate decision, §4.4 reports only the current 43-pair census rather than this trajectory, on the grounds that the 226- and 24-pair populations no longer describe the system as it stands; the full historical numbers remain on record in `docs/ANNOTATION_RESULTS.md` for a reader who wants the before/after rather than only the after.

News recency and outlet bias are a further, related pair of limits. 68.8% of dated articles in the conduct corpus were published in 2025 or 2026, against a claim side spanning fifteen years (§2.3.5), so a claim made early in that span has essentially no contemporaneous conduct evidence and is checked, if at all, against much later conduct; the asymmetric retrieval window in §3.8 is a direct response to this, not a design preference. Separately, one of the three search back-ends supplies roughly three-quarters of the corpus rather than the three-way balance the multi-back-end design was meant to produce (§2.3.5), so the conduct channel's topic and outlet mix is closer to what one ranking algorithm surfaces than to an unbiased sample of Vietnamese business news.

The ESG classifier's precision on this corpus is also not independently measured. ViDeBERTa-v3-ESG is applied as a published checkpoint, not fine-tuned or evaluated against a held-out Vietnamese label set (§2.1.2), because building one was outside this project's budget. Its behaviour on this corpus, fitted, as every such checkpoint is, to a training distribution assembled elsewhere, is therefore an empirical question this thesis does not close; every downstream stage inherits whatever precision and recall the classifier actually has on Vietnamese construction-sector disclosure, silently.

The self-verification (independence) guard, too, has been exercised for one issuer only. §3.8 drops a `supports` verdict back to a non-independent, dossier-visible flag whenever its evidence's source domain matches a curated list of the issuer's own web properties; that list is currently populated for the issuer this thesis's Methodology chapter worked through in detail, and it has never dropped an edge for any of the other four issuers, because every cited item for those four so far originates from a third-party outlet. The guard would not currently fire even if one of those four issuers' own domains appeared in a future conduct pool, since the curated list has no entry for them. The risk is latent, not realised, in the corpus this thesis evaluates. Contribution 2's independence claim (§1.3) is currently enforced structurally instead, by the `source_type` channel separation, which is real and does hold for all five issuers, rather than by this domain-level guard, which does not yet.

A crawl-only acquisition route creates a single point of failure, too. Report PDFs are located through one third-party disclosure mirror rather than fetched directly from each issuer or from the

State Securities Commission, so acquisition inherits whatever that mirror's own coverage, naming conventions and uptime happen to be; the two-year-field naming inconsistency documented in Chapter 2 traces to exactly this dependency.

Finally, the report-side extraction pipeline is a positional parser, not a layout-aware one, and this remains an argued design choice rather than a measured one. §2.2.3 exact-page-anchors every sentence at the cost of table reconstruction; the longest artefact this produces is a single 11,827-character text run that is a flattened table, not a sentence. No ablation in this thesis quantifies how much information, or how much noise, this trade-off costs against the layout-aware parser used for the much smaller reference corpus (§2.2.6); the choice is defended qualitatively, not empirically.

6. Conclusion

This thesis set out to test whether a temporal knowledge graph, fed by two structurally independent evidence channels, could make greenwashing in Vietnamese real-estate, construction and building-materials disclosure checkable rather than merely readable. The Delmas–Burbano definition of greenwashing, the intersection of what a company says and how it behaves, fixes the requirement directly: a system with access to only the company's own reports cannot in principle observe that intersection, whatever its architecture. This thesis builds the independent half deliberately, keeps it distinguishable from the claim half by a source-type stamp rather than fusing the two into one undifferentiated pool, and carries validity time and sentence-level provenance through every stage so a claim made in one year can be checked against conduct observed in another and traced back to the exact page it came from.

Instantiated end to end on five issuers, the system produces a schema-compliant, well-provenanced knowledge graph (10,624 nodes, 15,130 edges, 100% schema compliance, 97.78% round-trip KPI grounding) and, on the claims and evidence it actually cites, is confirmed by two independent domain experts working blind roughly two-thirds of the time, with near-perfect agreement between them. Both of those are genuine, externally-checkable results. Read against the aggregate stated in §5.2, however, they describe a narrow slice of the system's total output: 16 of 464 claims verdicted, concentrated overwhelmingly in one issuer, retrieved by lexical overlap rather than the graph-native indicator-axis join this design also specifies. The honest summary of this project is that the engineering and the schema are sound, the label-free evaluation instrument correctly detects a real regression and its repair, and the external validation is genuinely positive on what it covers; what remains unproven is that any of this generalises past a conduct channel roughly sixteen times smaller than the claim side it is meant to check.

What follows directly from that gap is not a new model or a new schema: it is more independent conduct evidence, since every quantity in §5.2 is a downstream consequence of how thin that channel currently is. Three further items are already specified by the design and require no new research: wiring the graph-native indicator-axis retrieval tier so it is actually consulted rather than only measured as coverage; driving the self-verification guard and its stopword list from `config/issuer_registry.json` per issuer rather than leaving it populated for one; and scaling report and news ingestion from five issuers to the full 115-issuer, sector-wide corpus this thesis already built and classified. None of these three is an open research question; each is an engineering task against an existing, tested design.

What the project got right, and would not change, is the opposite kind of decision. Returning `unverified` as a first-class outcome, rather than forcing every claim to a verdict, is the correct response to a conduct channel this thin, and is more honest than the alternative of a confident wrong answer. Refusing to emit a greenwashing score or label, in the total absence of ground truth for this or any Vietnamese corpus, is the same discipline applied one level up, and it is what makes the label-free evaluation instrument in this thesis, rather than a spurious accuracy number, the right kind of evidence to have built. A system that says less, but says it traceably and only when it can, is the contribution this thesis makes to a problem that has so far mostly been approached the other way.

A. Full Record Schema

Table A.1 lists every field of the labelled sentence corpus record schema (§2.1.1). The first seven fields — the five core fields shown in Table 2.1, plus `scores` and `esg` from the ESG classifier — are present in both channels; the remaining eight, from `ticker/company` onward, exist only in the news channel.

Table A.1: Full record schema of the labelled sentence corpus.

Field	Type	Meaning
`source_pdf`	string	Source document identifier (report file name, or ticker + domain + hash for an article)
`page`	int	1-based page of the source PDF (always 1 for an article)
`sentence_index`	int	Position of the sentence within that page
`text`	string	The sentence, verbatim, NFC-normalised
`labels`	list	Pillars whose score reaches the tag threshold; may be empty or hold several
`scores`	object	Four sigmoid scores: Neutral, Environmental, Social, Governance
`esg`	bool	Binary ESG relevance, decided by the Neutral score (§2.2.4)
`ticker`, `company`	string	Issuer the article was retrieved for
`url`, `source_domain`, `title`	string	Article location, publisher and headline
`publish_date`, `date_crawled`	date	Stated publication date; retrieval timestamp
`channel`, `query`	string	Search back-end used, and the query that returned the article
`matched_terms`, `company_mentioned`	list, bool	Which issuer identifiers occur in the sentence

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